

UNIVERSITE PARIS CITE

MASTER THESIS

The Effect of Scaling Down Minicolumn Size on SNN Associative Memory Performance

Author:
Ferdinand BUJANOWSKI

Supervisor:
Dr. Pawel HERMAN

*A thesis submitted in fulfillment of the requirements
for the degree of Master*

in the

Computational Brain Science Lab
School of Electrical Engineering and Computer Science, KTH Royal Institute
of Technology

May 20, 2026

Declaration of Originality

I, Ferdinand BUJANOWSKI, declare that this thesis titled, “The Effect of Scaling Down Minicolumn Size on SNN Associative Memory Performance” is an original work, going beyond what has already been published in the literature. Some slight changes have been made to the research questions and working hypotheses compared to what is written in the pre-registration from January of this year (shown in appendix **B**): While the question of the effect of scaling down the number of simulated neurons per minicolumns on attractor memory performance and stability stayed the same, the proposed transfer of a multi-layer Poisson-spiking neural network into a integrate-and-fire SNN of comparable function (image reconstruction) was abandoned, due to the large scale of the original network. Instead, a smaller SNN was built from scratch, and subsequently trained on patterns not representing any real-world data. While the network scale was largely reduced, the conclusions drawn from the experimental evidence presented in this thesis stayed comparable to the hoped-for results originally described in the pre-registration.

Declaration of Contribution

The following people contributed to the creation this thesis: Ferdinand BUJANOWSKI, FB; Pawel HERMAN, PH (principal supervisor); Anders LANSNER, AL (supervisor); Florian FIEBIG, FF (researcher at host laboratory); Angeliki PAPADIMITRIOU, AP (PhD student at host laboratory):

- Defining the scientific question: PH, FB
- Proposing relevant literature: PH, AL
- Reviewing literature: FB
- Programming the model, writing code for experiments & data analysis: FB
- Choice of methodology: AL, PH, FB
- Interpreting results / model behavior: FB, AL, PH, FF
- Providing feedback on model parameter tuning PH, AL, FF, AP
- Running model simulations & data analysis code: FB
- Plotting model data: FB
- Computing result statistics: FB
- Writing the thesis: FB
- Proof-reading & providing feedback on the thesis: AL, PH

UNIVERSITE PARIS CITE

Abstract

Computational Brain Science Lab

School of Electrical Engineering and Computer Science, KTH Royal Institute of Technology

Master

The Effect of Scaling Down Minicolumn Size on SNN Associative Memory Performance

by Ferdinand BUJANOWSKI

Artificial neural networks - computational models inspired by the organization of nerve tissue into neurons and synapses - have long revolutionized both industrial as well as academic sectors, due to their vast capabilities in learning patterns from data without explicit instructions. A central question when designing ANNs is that of their implemented degree of biological plausibility. While certain kinds of ANNs widely used in industrial and scientific contexts present strong simplifications and abstractions from their biological counterpart, the question of implementing biological principles into ANNs is not trivial. Biologically plausible ANNs can serve as valuable in silico models of the brain, its circuitry as well as larger cognitive phenomena like attention, action selection and memory. When modeling biological neural networks, a tradeoff between network complexity and computational cost as well as execution time exists. It is the purpose of this thesis to study and assess a number of guidelines that can be followed when designing artificial spiking neural networks of attractor memory, in order to minimize the amount of simulated neurons while keeping memory function largely intact...

Contents

Declaration of Originality	iii
Declaration of Contribution	v
Abstract	vii
1 Introduction	1
1.1 Wider context and scope of the thesis	1
1.2 Research questions & hypotheses	2
2 Background	3
2.1 Recurrent neural networks & cell assemblies	3
2.2 The mammalian neocortex	4
3 Methods	5
3.1 Network setup	5
3.2 Network equations	7
3.2.1 Neuron model	7
3.2.2 Synapse model	8
3.2.3 Spike-based BCPNN plasticity	9
3.2.4 Stimulation protocol	11
3.3 Model tuning & parameter scaling laws	12
3.4 Model Implementation	14
3.5 Pattern generation & performance evaluation	15
4 Results	17
4.1 Simulations with orthogonal patterns	17
4.2 Simulations with 20 random patterns	19
4.3 Reducing N_{PYR}	22
5 Discussion	25
A Supplementary Figures	27
B Internship Pre-Registration	31
Bibliography	35

Chapter 1

Introduction

1.1 Wider context and scope of the thesis

Artificial neural networks (ANNs) are the backbone of many modern-day artificial intelligence algorithms and applications and have long revolutionized many tech industries as well as sectors such as healthcare and academia, due to their incomparable capabilities to implicitly learn to perform tasks that are otherwise too complex to possibly write explicit instructions for. While originally inspired by the organization of the brain into neurons which communicate via synapses, many ANNs finding large-scale use in modern machine learning applications such as data mining, computer vision, and most recently natural language processing, are far removed from the biochemical and structural complexity that makes up real biological nervous systems. However, in the academic field of computational neuroscience there is an ongoing effort in developing neural networks with more biologically plausible architectures as well as local (Hebbian) learning rules, compared to global learning rules such as the backpropagation algorithm utilized in deep neural networks (DNNs) that excel in many of the aforementioned tasks and sectors (Werbos, 1990). The interest in more biologically plausible ANNs is twofold: One motivation behind this effort is the fact that the brain is highly energy-efficient while being notoriously good at learning patterns in the environment without explicit instruction and without much repetition. And while global (and thus less biologically plausible) learning rules might be mathematically more sound and error-proof compared to local learning at the synaptic scale, DNNs require a lot more time and thus computational resources to successfully learn. Thus, deciphering the computational paradigms underlying learning and memory in the brain might make large-scale machine learning applications more cost- and energy-efficient in the long run. Another interest driving research into biologically plausible neural networks is that they can provide an excellent tool for testing hypotheses about causal relationships between structure and function in the brain, which would be impossible to test *in vivo*. This can improve our understanding of the healthy and dysfunctional brain and support development of better diagnosis and treatment of neurological diseases like Alzheimer and Parkinson as well as neuropsychiatric conditions like depression and schizophrenia.

A subclass of biologically inspired neuromorphic models is that of spiking neural networks, or SNNs. These models work by simulating synaptic communications between computational units - cortical minicolumns or single neurons depending on the type of SNN - via the emission of action potentials, or spikes. A benchmark cognitive ability of the brain, which SNNs are able to reproduce, is the learning and maintaining of so-called associative memory (Hebb, 1949, Lansner, 2003), also called attractor (Hopfield, 1982) or content-addressable memory. This refers to memory that can be (re)activated by exposing it to distorted or partial input corresponding to

the memorized object, or to input corresponding to an associated object (Ravichandran, Lansner, and Herman, 2025).

Biologically inspired ANNs often follow a structural organization found in the neocortex into hypercolumns and minicolumns. Minicolumns are widely considered the smallest units of neural computation, and consist of roughly 80 to 100 neurons that are densely connected among each other, thus displaying strongly correlated spiking (Mountcastle, 1997). Many biologically inspired SNNs thus do not model each neuron separately, but rather model at the scale of minicolumns and hypercolumns, describing each minicolumn as a population firing rate dependent on those of the other network columns. While these networks are capable of forming stable associative memories (Ravichandran, Lansner, and Herman, 2024), an unexplored question is whether these findings can be reproduced in SNNs modeled at the scale of individual neurons. Modeling each neuron individually largely increases the computational cost of simulating such networks compared to the population rate-based ones, yet they are more biologically plausible given their greater level of biophysical detail. Another question worth exploring is thus how the amount of simulated individual neurons per minicolumn in associative memory SNNs can be reduced without affecting overall memory performance of the network.

1.2 Research questions & hypotheses

Two central research questions that are the focus of this thesis are the following: First, can spiking neural networks modeled at the minicolumn level capable of learning associative memories be reformulated as models defining each neuron separately, while keeping network learning as well as subsequent associative memory capabilities intact? And 2) Is it possible to reduce the amount of neurons simulated for each minicolumn in such a network without affecting memory performance?

The two working hypotheses tested in this thesis are thus the following:

1. *It is possible to build spiking neural networks, simulating individual neurons rather than minicolumn neuron populations, that are capable of learning stable associative memories through Hebbian learning rules.*
2. *When reducing the amount of neurons simulated per minicolumn in such a network, attractor memory performance will reduce; there exists however a set of necessary and sufficient parameter scaling operations that will minimize memory performance loss throughout this reduction process.*

In the following, a computational model of a spiking neural network will be developed, on which these hypotheses will be tested. In order to do so, a "control" network, having a number of neurons per minicolumn that is comparable to previous computational models, will be trained on both orthogonal as well as overlapping, randomly generated patterns, and associative memory performance will be evaluated. Next, several simulation batches will be run on different model versions with less and less neurons simulated per minicolumn, and a set of potential model retuning operations will be critically assessed with regards to their respective effect on total network memory performance.

Chapter 2

Background

2.1 Recurrent neural networks & cell assemblies

The brain is composed of nerve cells called neurons, which transmit electrochemical signals called action potentials via connections called synapses. Neurons emit action potentials in response to their interior electrochemical potential, more specifically when this potential goes over a certain voltage threshold. At a given synapse, when the pre-synaptic neuron emits an action potential (also called "firing"), it triggers an increase or decrease in the post-synaptic neuron's potential (see Section 2.2). The amount of this transient change in potential is determined by the strength of the synaptic connection between the two neurons. Networks of neurons can generally be split into two categories based on their connectivity: feed-forward, meaning that synaptic connections do not form any closed loops of neurons, versus recurrent, meaning that all neurons can be synaptically connected to any other neuron, allowing for the existence of loops. While the exact make up of connectivity in the brain varies from region to region, it is generally assumed that the cerebral cortex (see Section 2.2) is recurrent in nature (Douglas et al., 1995). Given that in a recurrent neural network, the change of firing activity of each neuron can be described as a function of momentary firing activities of all pre-synaptically connected other neurons, it can mathematically be described as a dynamical system (Meng et al., 2022), allowing for the existence of so-called attractors. An attractor is a state in a dynamical system that is stable over time, and that the system naturally converges to when in close proximity (i.e. in its *basin of attraction*). In the context of neural networks, it refers to a state where all neurons fire at fixed rates, and where small perturbations of any single neuron's firing lead it back to its respective stable firing rate.

It has long been theorized that the brain stores information and forms memories not inside single cells, but rather through patterns of activity of subsets of neurons, called cell assemblies. These assemblies were proposed by Donald Hebb to form through a biochemical process in which the synaptic strength between two or more neurons firing at the same time would become stronger (Hebb, 1949), thus theoretically leading to the emergence of stable attractor states like those mentioned above. This kind of coincidence-driven learning of cell assemblies on the local scale of each synapse (i.e. the only information a synapse uses to update its strength are the activities of both pre- and postsynaptic neurons) is referred to in the literature as Hebbian (Mayorquin, 2022, Tsien, 2000, Caporale and Dan, 2008). Computational models of cortical recurrent neural networks have shown that dynamically stable attractor states can be formed through Hebbian learning rules (Lundqvist, 2013, Hopfield, 1982, Sandberg, Tegner, and Lansner, 2003).

Due to the dynamical properties of their basin of attraction, cell assemblies representing memorized previous input stimuli show interesting properties. Namely,

they are able to reconstruct full memories from partial or distorted input cues. In addition, it has recently been shown that recurrent neural networks trained on benchmark machine learning datasets like MNIST (Lecun et al., 1998) can extract underlying prototypes of the data without explicitly being instructed to do so (Ravichandran, Lansner, and Herman, 2024).

2.2 The mammalian neocortex

The cerebrum is one of 7 main areas of the brain, which along with the spinal chord makes up the central nervous system. The cerebrum is the upper and outermost portion of the brain, and can generally be split into white matter in its center, and gray matter at its surface. While the former is mainly composed of myelinated axons, the latter, also called the cerebral cortex, is composed mainly of neural cell bodies and is considered to be where higher-order cognitive processes take place (Mayorquin, 2022; Lodato and Arlotta, 2015). Generally speaking, cortical neurons can be either excitatory (glutamatergic) or inhibitory (GABA-ergic), with a ratio of roughly 80% excitatory to 20% inhibitory (Mayorquin, 2022, DeFelipe and Fariñas, 1992, Sukenik et al., 2021). Through emission of action potentials, excitatory neurons depolarize neurons that they are pre-synaptically connected to, meaning that the respective post-synaptic neuron's membrane potential is slightly raised, whereas inhibitory neurons do the opposite (lowering the post-synaptic neuron's membrane potential) which is called hyperpolarization. Another structural characteristic of the cerebral cortex is that of its lamination; when observing cortical cell composition as well as synaptic connectivity, along its radial axis, distinct layers can be made out. In the neocortex, a portion of the cortex phylogenetically younger than its counterpart, the allocortex, there are six distinguishable cortical layers, numbered I to VI (Thomson and Bannister, 2003). Recordings of neural activity and connectivity in the neocortex suggest that distinct computational units in the neocortex follow a columnar architecture (Mountcastle, 1957, Mountcastle, 1997), where each unit, or column is defined by an area of shared input and increased recurrent connectivity (Bastos et al., 2012, Douglas et al., 1995) perpendicular to the lamination axis, thus spanning all cortical layers. In addition, it has been proposed that local groups of columns, defined by the extent of their shared inhibition and thus participating in local competition, together encode input features, where each column in the group represents one of several possible values that this input feature could take (Carandini and Heeger, 2012). Following the naming convention common in computational modeling papers, these two levels of columnar organization will henceforth be referred to as hypercolumns and minicolumns, respectively. Combining the findings from last section that groups of neurons can form assemblies through coactive spiking as a result of Hebbian learning with the idea of only one cortical minicolumn of each hypercolumn can be active due to local competition, it can be concluded that associative memory cell assemblies in the cerebral cortex should be distributed across several hypercolumns.

Chapter 3

Methods

3.1 Network setup

The spiking neural network developed in this thesis is composed of a single recurrent layer of Layer II/III (see Section 2.2) pyramidal (PYR) cells (DeFelipe and Fariñas, 1992) as well as basket (BA) cells (McBain, 2012), where pyramidal cells project synapses onto each other without autapse, meaning that no neuron can have a synapse with itself. In addition, PYR neurons project synapses onto basket cells, and vice-versa. The simulated recurrent layer follows the structural organization found in the mammalian neocortex of hypercolumns composed of minicolumns, which are in turn composed of several PYR and BA cells. Specifically, the network simulated for this thesis is composed of $N_{HC} = 9$ hypercolumns of $N_{MC} = 9$ minicolumns each. Each minicolumn consists of $N_{PYR} = 30$ pyramidal as well as $N_{BA} = 4$ basket cells, making the total amount of simulated cells equal to $N_{total} = 2754$.

Minicolumns being considered the smallest units of neural computation, all pyramidal cells within the same minicolumn are considered to behave "as one" in terms of their activity and overall spiking dynamics. This means that inside a given MC (and without external inhibition), one pyramidal neuron spiking at an elevated rate generally speaking activates its neighbor pyramidal neurons within the same MC, leading to a positive feedback loop that "turns on" the entire minicolumn into a persistent state of heightened spiking activity. To observe this behavior *in silico*, the weights of the intra-MC recurrent PYR-to-PYR synapses (which are non-plastic, see Section 3.2.3) are initialized to be very strong (Table 3.1), roughly 50% stronger than the maximum positive weight attainable through the BCPNN learning rule (see Section 4). Recurrent synapses inside each minicolumn are drawn randomly with a probability of $p_c^{IntraMC} = 25\%$ following previous modeling studies (Fiebig, Herman, and Lansner, 2020, Lundqvist, Herman, and Lansner, 2011). In the literature it is stated that the average number of incoming synapses onto a pyramidal neuron from other pyramidal neurons of the same minicolumn is in the order of around ten (Lundqvist, Herman, and Lansner, 2011, Thomson et al., 2002), which the here chosen connection probability satisfies ($0.25 * N_{PYR} = 7.5$).

In addition to $N_{PYR} \times N_{MC} = 270$ pyramidal neurons, each hypercolumn has a total of $N_{BA} \times N_{MC} = 36$ basket cells. Synapses from pyramidal to basket as well as basket to pyramidal cells are randomly drawn with connection probability $p_c^{PB} = 70\%$. Basket cells have considerable inhibitory power over pyramidal neurons (see Table 3.3). Given this strong inhibition as well as the local within-hypercolumn connectivity between PYR and BA cells, the latter engage their respective hypercolumns in so-called local winner-take-all (WTA) dynamics between minicolumns (Binzegger, Douglas, and Martin, 2009): If two or more minicolumns of the same HC are active (meaning that they have heightened persistent spiking activity) at the same time, basket cells will start firing, which silences all MCs of that given HC for

a short time. Of all the previously active MCs, the one that had the strongest activity (usually a MC participating in a currently active attractor spanning the entire network) will recover more quickly than the rest, effectively making it the "winner" inside that given HC.

In this model, pyramidal cells of a given minicolumn do not project recurrent synapses onto other pyramidal cells of the same hypercolumn but different minicolumns, meaning that competition between two or more minicolumns inside the same hypercolumn is solely driven by basket cell spiking deploying local WTA dynamics. Recurrent synapses between pyramidal cells of different hypercolumns do exist, and it is precisely those synapses that are plastic according to the BCPNN learning rule (Section 3.2.3). While pyramidal cells in the brain can only be excitatory, in this model they can also establish inhibitory synapses in function of synaptic weights learnt by the BCPNN rule, implicitly modeling di-synaptic inhibition through intermediate inhibitory neurons (see Section 3.2.3).

While in real nervous tissue, conduction delay between the emission of an action potential to the release of neurotransmitters at the synapse depends on the physical distance between two given neurons as well as other parameters such as axon myelination (Seidl, 2014), this model simplifies synaptic conduction delay by randomly sampling delay times from two distinct distributions: The first category of conduction delays concerns all intra-hypercolumn synapses, i.e. all recurrent intra-MC PYR-PYR synapses as well as all PYR-BA and BA-PYR synapses. For each of these synapses, conduction delay is uniformly sampled from distribution t_{delay}^{short} (between 1.5 and 1.7 milliseconds). The second category of conduction delay concerns the more long-range inter-hypercolumn recurrent PYR-to-PYR synapses. Each of these synapses' conduction delay is thus uniformly sampled from t_{delay}^{long} (between 15 and 25 milliseconds). Assuming linear increase in conduction delay with growing physical distance, each neuron is thus considered to be roughly ten times closer to all neurons within its own hypercolumn than it is to neurons of different hypercolumns.

Parameter	Symbol	Value
Simulated Hypercolumns	N_{HC}	9
Stimulated Minicolumns	N_{MC}	9
Pyramidal cells per MC*	N_{PYR}	30
Basket cells per MC*	N_{BA}	4
Total simulated cells*	N_{total}	2754
Intra-MC connection weight	$w^{IntraMC}$	$3 \times w_{gain}^{syn}$
Intra-MC connection probability*	$p_c^{IntraMC}$	0.25
Inter-HC connection probability*	$p_c^{InterHC}$	0.38
PYR-BA connection probability	p_c^{PB}	0.7
BA-PYR connection probability	p_c^{BP}	0.7
Short-range conduction delay	t_{delay}^{short}	[1.5 ms, 1.7 ms]
Long-range conduction delay	t_{delay}^{long}	[15 ms, 25 ms]

TABLE 3.1: Network architecture, connectivity & conduction delay parameters. Parameters marked with * will subsequently be modified during network size reduction tests (see Sections 3.3 and 4.3 as well as Table 3.3).

3.2 Network equations

3.2.1 Neuron model

Pyramidal neurons were implemented following the adaptive exponential integrate-and-fire (AdEx) model (Brette and Gerstner, 2005), whereas basket neurons were implemented using the slightly simpler leaky integrate-and-fire (LIF) equation (Lapique, 1907). Following previous model implementations, the AdEx model equation was slightly modified for BCPNN compatibility, adding an intrinsic excitability current I_β , a crucial step for adapting the BCPNN learning rule into the spiking regime (Tully, Hennig, and Lansner, 2014). For a given PYR neuron, the rate of change of its membrane voltage V_m and adaptation current I_w are described by the following equations:

$$C_m \frac{dV_m}{dt} = -g_L(V_m - E_L) + g_L \Delta_T e^{\frac{V_m - V_t}{\Delta_T}} - I_w + I_\beta - I_{syn} - I_{ext} \quad (3.1)$$

$$\frac{dI_w}{dt} = \frac{I_w}{\tau_{I_w}} + bS \quad (3.2)$$

As Equation 3.1 shows, the momentary change of membrane voltage can be expressed as the sum of currents passing through the membrane at a given time, divided by the total membrane capacitance C_m . Current contributions include the leak current driving the membrane potential towards its resting potential E_L , the intrinsic excitability current I_β , total incoming synaptic current I_{syn} , external input current I_{ext} , and lastly an exponential voltage dependent activation term as well the adaptation current I_w . The latter gets upregulated by b with each spike of the neuron (triggered by the membrane potential exceeding the threshold V_t , at which time it is reset to V_r) and recovers with time constant τ_{I_w} . The spike indication variable S is equal to 1 at the moment of a spike, otherwise 0 (see Section 3.4). Pyramidal neurons receive synaptic currents I_{syn} through conductance-based glutamatergic and GABA-ergic synapses. Glutamatergic synapses contain both AMPA and NMDA receptor gated channels with respectively fast and slow conductance decay dynamics (Lisman, 2017), whereas GABA-ergic synapses modeled here only feature receptor gated channels with fast dynamics (GABA-A). Synaptic current I_{syn} is defined as follows for a given pyramidal neuron j :

$$\begin{aligned} I_{synj} &= I_j^{AMPA} + I_j^{NMDA} + I_j^{GABA} \\ &= \sum_{i_{PYR}} g_{ij}^{AMPA} (V_m - E^{AMPA}) \\ &\quad + \sum_{i_{PYR}} g_{ij}^{NMDA} (V_m - E^{NMDA}) \\ &\quad + (\sum_{i_{PYR}} g_{ij}^{GABA} + \sum_{i_{BA}} g_{ij}^{BP}) (V_m - E^{GABA}) \end{aligned} \quad (3.3)$$

For basket cells, being modeled without adaptation, intrinsic excitability or external input currents, their equation of momentary voltage change is simpler than that of PYR neurons:

$$C_m \frac{dV_m}{dt} = -g_L(V_m - E_L) - \sum_{i_{PYR}} g_{ij}^{PB} (V_m - E^{AMPA}) \quad (3.4)$$

Parameter	Symbol	Value
Adaptation current	b	50 pA
Adaptation time constant	τ_{Iw}	280 ms
Membrane capacitance	C_m	280 pF
Leak reversal potential	E_L	-70 mV
Leak conductance	g_L	14 nS
Upstroke slope factor	Δ_T	3 mV
Spike threshold	V_t	-55 mV
Spike reset potential	V_r	-80 mV
Membrane refractory period	t_{ref}	5 ms
AMPA reversal potential	E_{AMPA}	0 mV
NMDA reversal potential	E_{NMDA}	0 mV
GABA reversal potential	E_{GABA}	-75 mV

TABLE 3.2: Neuron Model Parameters

3.2.2 Synapse model

Incoming synaptic current from glutamatergic synapses drives pyramidal membrane voltage towards $E^{AMPA} = E^{NMDA}$, whereas inhibitory current from GABAergic synapses drives it towards E^{GABA} . Incoming presynaptic spikes evoke transient changes in synaptic conductance g_{ij} following exponential decay with time constant τ^{syn} depending on the synapse type, mediated by the synaptic variable H^{syn} :

$$\frac{dH_{ij}^{syn}}{dt} = \frac{-H_{ij}^{syn}}{\tau^{syn}} + S_i(1 - H_{ij}^{syn}) \quad (3.5)$$

At the time of a presynaptic spike (transmitted to the synapse with delay t_{ij}), S_i is equal to 1, otherwise 0. Thus, the momentary synaptic conductance g_{ij} can be expressed as the following (time indices omitted for simplicity):

$$g_{ij}^{syn} = x_{ij}^{dep} w_{ij}^{syn} H_{ij}^{syn} \quad (3.6)$$

w_{ij}^{syn} is the synaptic weight representing the peak amplitude of the conductance transient, which is learnt by the spike-based BCPNN rule (3.2.3). In addition, glutamatergic synapses are subject to synaptic depression, following the Tsodyks-Markram formalism (Tsodyks and Markram, 1997), modeled after the idea that each spike releasing neurotransmitters into the synaptic cleft reduces the amount of neurotransmitters that can be released in subsequent spikes, thus affecting the maximum amplitude of transient increase in synaptic conductance. This synaptic depression recovers with time constant τ_{ref} , modeling the re-uptake of neurotransmitters.

$$\frac{dx_{ij}^{dep}}{dt} = \frac{1 - x}{\tau_{rec}} - S_i U x_{ij}^{dep} \quad (3.7)$$

PYR neurons project synapses onto BA cells and vice-versa. Just like for PYR-PYR recurrent connections, each incoming spike at PYR-BA or BA-PYR synapses triggers a transient increase in respective synaptic conductances g^{PB} and g^{BP} , mediated by exponentially decaying synaptic variables H^{PB} and H^{BP} (see equation 3.5). For a pair of pyramidal and basket cells i and j , these synaptic conductances can thus

be respectively expressed as follows:

$$g_{ij}^{PB} = G_{PB} H_{ij}^{PB} \quad (3.8)$$

$$g_{ji}^{BP} = G_{BP} H_{ji}^{BP} \quad (3.9)$$

Pyramidal-to-basket synapses being excitatory in nature, H^{PB} decays with time constant $\tau^{PB} = \tau^{AMPA}$, while H^{BP} decays with $\tau^{BP} = \tau^{GABA}$, as basket cells are inhibitory i.e. GABA-ergic in nature. G_{PB} and G_{BP} are constant amounts of conductance, by which g^{PB} and g^{BP} get thus respectively increased at the time of each respective spike (see Table 3.3).

Parameter	Symbol	Value
Utilisation factor	U	0.2
Depression time constant	τ_{rec}	280 ms
AMPA synaptic time constant	τ^{AMPA}	5 ms
NMDA synaptic time constant	τ^{NMDA}	20 ms
GABA synaptic time constant	τ^{GABA}	5 ms
Training plasticity	$\kappa_{training}$	1
Recall plasticity	κ_{recall}	0
BCPNN AMPA gain*	w_{gain}^{AMPA}	0.61 nS
BCPNN NMDA gain*	w_{gain}^{NMDA}	0.02 nS
BCPNN GABA gain*	w_{gain}^{GABA}	1.97 nS
PYR-BA synaptic conductance*	G_{PB}	0.53 nS
BA-PYR synaptic conductance	G_{BP}	16 nS
BCPNN bias current gain	β_{gain}	40 pA
BCPNN lowest firing rate	f_{min}	0.5 Hz
BCPNN highest firing rate	f_{max}	50 Hz
BCPNN lowest probability	ϵ	0.01
BCPNN spike event duration	t_{spike}	0.1 ms
Z trace time constant	τ_Z	20 ms
P trace time constant	τ_P	5 s

TABLE 3.3: Synapse & BCPNN Plasticity Parameters

3.2.3 Spike-based BCPNN plasticity

Each recurrent pyramidal-to-pyramidal synapse possesses an individual strength, or weight, which modulates the total amplitude of transient rise in conductance (g^{syn}) depolarizing or hyperpolarizing the synapse's respective postsynaptic neuron at the time of a presynaptic spike. For synapses that are between two neurons of the same functional unit (or minicolumn, see Section 3.1), this weight parameter is fixed, or non-plastic ($w_{ij}^{syn} = w^{IntraMC}$, see Table 3.1), whereas the weights of all pyramidal-to-pyramidal synapses between neurons of different mini- and hypercolumns get updated over time according to the BCPNN learning rule (Chrysanthidis et al., 2025, Tully, Hennig, and Lansner, 2014, Lansner and Ekeberg, 1989). The BCPNN learning rule was chosen for this model mainly for the reason that many computational models of similar nature used as references for this model (Section 3.4) had also used it.

In addition, this learning rule appears to perform exceptionally well in pattern memorization tasks compared to other Hebbian learning rules (Lansner, Ravichandran, and Herman, 2025).

At each of these latter synapses, pre- as well as postsynaptic spiking activities are estimated and averaged over time using biologically plausible, local memory traces. First, the so-called Z-traces compute exponential moving averages (EMAs) using their respective spike trains S_i and S_j :

$$\tau_{Z_i} \frac{dZ_i}{dt} = \frac{S_i}{f_{max} t_{spike}} - Z_i + \epsilon, \tau_{Z_j} \frac{dZ_j}{dt} = \frac{S_j}{f_{max} t_{spike}} - Z_j + \epsilon \quad (3.10)$$

The amount that the Z-traces increase at each spike event gets scaled by the maximum expected firing frequency f_{max} and the spike timing duration t_{spike} , which is equal to the simulation time step t_{sim} . Thanks to these scaling operation, the Z-trace mean over time converges to 1 when the respective pre- or post-synaptic neuron fires at f_{max} . The smallest possible value that the Z-traces can take is ϵ . While there is ongoing discussion in the literature regarding what the value of ϵ should be and how its choice affects network spiking dynamics (Mayorquin, 2022), in this thesis it is set to $\frac{f_{min}}{f_{max}}$. Essentially, the Z-traces allow for translating neuron firing rates into probability values, with the maximum probability being reached when a given neuron's firing rate is equal to f_{max} , while being equal to ϵ when the firing rate is equal to f_{min} .

While at glutamatergic synapses, AMPA current contributes most to the amplitude of excitatory post-synaptic potentials compared to NMDA current (given that $w_{gain}^{AMPA} > w_{gain}^{NMDA}$), literature suggests that synaptic plasticity is mostly driven through NMDA current (Hunt and Castillo, 2012). For that reason, in this model $\tau_{Z_i} = \tau_{Z_j} = \tau^{NMDA}$.

From the pre- and postsynaptic Z-traces, the P-traces of pre-, post- as well as co-activation are estimated as follows:

$$\tau_P \frac{dP_i}{dt} = \kappa(Z_i - P_i), \tau_P \frac{dP_j}{dt} = \kappa(Z_j - P_j), \tau_P \frac{dP_{ij}}{dt} = \kappa(Z_i Z_j - P_{ij}) \quad (3.11)$$

The parameter κ toggles plasticity in the model. During training phases (see Section 4), it is set to $\kappa = \kappa_{training} = 1$, while in phases of recall it is equal to $\kappa = \kappa_{recall} = 0$, freezing all P-traces, and thus all weights and biases. The latter allows for testing memory capacities without forgetting previously memorized patterns or learning wrong associations in distortion correction tasks. It should be mentioned that while in this model κ serves only as a binary flag of plasticity, in other model implementations it can take more values than 0 and 1.

Finally, the P-traces P_i , P_j and P_{ij} are used to compute the synaptic weights w_{ij} as well as each neuron's intrinsic excitability β_j :

$$w_{ij}^{syn} = w_{gain}^{syn} \log\left(\frac{P_{ij}}{P_i P_j}\right), I_{\beta_j} = \beta_{gain} \log(P_j) \quad (3.12)$$

The logarithm of the ratio between the co-activation P-trace and the product of pre- and postsynaptic P-traces makes it so that the sign of the resulting weight changes in function of the correlation between pre- and postsynaptic spike trains: If the two spike trains are correlated, meaning that both neurons spike within overlapping time windows, then $P_{ij} > P_i P_j$, resulting in a positive weight (since the ratio will be > 1). If the spike trains are not correlated, then $P_{ij} \approx P_i P_j$, making the weight

$w_{ij} \approx \log(1) = 0$. Lastly, if the two spike trains are negatively correlated, meaning that one neuron only fires when the other is silent and vice-versa, the weight will become negative, since the ratio of P-traces will be < 1 ($P_{ij} < P_i P_j$). In this last case, the negative weight will be interpreted as the positive weight of a GABA-ergic synapse, and the synapse will switch from deploying glutamatergic current to its post-synaptic neuron through conductances g_{ij}^{AMPA} and g_{ij}^{NMDA} to GABA-ergic current through g_{ij}^{GABA} (see Equation 3.6). While pyramidal neurons in the brain only project glutamatergic synapses (DeFelipe and Fariñas, 1992), in this model they can have both glutamatergic as well as GABA-ergic synapses. This is simply a simulation shorthand for including di-synaptic inhibition without having to explicitly simulate a third neuron type along with pyramidal and basket cells. The GABA-ergic conductances are thus interpreted as being mediated by dendritic targeting regular spiking non-pyramidal (RSNP) cells, such as, for example, double-bouquet cells (Krimer et al., 2005, Kelsom and Lu, 2013). In fact, a previous study has shown that this method of allowing for excitatory as well as inhibitory synapses within the same recurrent layer of pyramidal-to-pyramidal synapses is mathematically equivalent to explicitly simulating RSNP cells alongside purely glutamatergic pyramidal cells (Chrysanthidis, Fiebig, and Lansner, 2018).

3.2.4 Stimulation protocol

The external current I_{ext} to each PYR neuron is made up of excitatory (AMPA) and inhibitory (GABA) currents whose respective conductances are controlled by three independent Poisson spike generators (Buesing et al., 2011): P_+ , P_- and P_{stim} . The first two generators represent excitatory and inhibitory background noise respectively, while the third represents an explicit additional excitation used for encoding memory in the network, thought to originate from other cortical layers or subcortical regions not explicitly modeled here (see Section 2.2).

$$\frac{dg_+}{dt} = \frac{-g_+}{\tau_{AMPA}} + G_{bg} S_{P_+}, \quad \frac{dg_-}{dt} = \frac{-g_-}{\tau_{GABA}} + G_{bg} S_{P_-} \quad (3.13)$$

$$\frac{dg_{stim}}{dt} = \frac{-g_{stim}}{\tau_{AMPA}} + b_{on} G_{stim} S_{P_{stim}} \quad (3.14)$$

S_{P_+} , S_{P_-} and $S_{P_{stim}}$ are equal to 1 whenever their respective Poisson generator generates a spike event, otherwise 0. The flag b_{on} is equal to 1 for a given neuron during training phases when the respective minicolumn of said neuron is participating in a new pattern. The total external current I_{ext} is equal to:

$$I_{ext} = (g_+ + g_{stim})(V_m - E^{AMPA}) + g_-(V_m - E^{GABA}) \quad (3.15)$$

Upon creation of the network, and before any memorization or recall simulations have been run, it should be in a "naive" state where all inter-HC weights are equal to 0. To achieve this state, all pre- and postsynaptic P-traces can be set to ϵ reflecting the lowest possible probability estimate of previous spiking activity. Knowing that $w_{ij} = 0$ if and only if $P_{ij} = P_i P_j$ (see Equation 3.12), the co-activation P traces of all plastic synapses can thus be set to ϵ^2 .

In subsequent model simulations, a distinction will be made between two simulation phases: the *training phase* as well as a the *recurrent or recall phase*. During the training phase, which is used for storing new patterns in the network, synaptic plasticity for all plastic (inter-HC) synapses is turned on ($\kappa=1$). This means that the P-traces of those synapses get continuously updated in function of respective

Parameter	Symbol	Value
Background noise input rate	r_{bg}	550 Hz
Background conductance	G_{bg}	1.5 nS
Stimulation rate	r_{stim}	600 Hz
Cue stimulus conductance	G_{stim}	1.5 nS
Training stimulus duration	t_{stim}	100 ms
Cue stimulus duration	t_{cue}	50 ms
Interstimulus interval	t_{ISI}	150 ms
Simulation time step	t_{sim}	0.1 ms

TABLE 3.4: Stimulation Protocol Parameters

pre- and postsynaptic spike trains, affecting the respective synaptic weights (see Section 3.2.3). However, in order to avoid potential attractor interferences from already stored patterns, recurrence is usually turned off during the input phase. This means that, while plastic synapses get actively strengthened or weakened over time, they are not actually allowed to convey post-synaptic potentials onto their respective post-synaptic neurons. It is only after training, during the recall phase that all synapses get turned on (while plasticity κ gets set to 0), allowing for network activity to converge towards potential attractor states learnt during the input phase.

During training, one pattern after the other will be "revealed" to the network, by setting the b_{on} flags of all pyramidal neurons participating in said pattern to 1, causing each of those neurons' P_{stim} Poisson generators to induce membrane potential depolarizations, leading to increased spiking activity. At the same time, all other neurons (i.e. those not participating in the currently trained pattern) can either be left to spike at baseline firing rate, or they can be subjected to inhibitory current driven by P_{stim} by setting $b_{on} = -1$, effectively turning I_{ext} into a purely inhibitory current. The latter allows for a more optimal learning outcome, as it reduces the risk of the network learning positive weights between attractor and a non-attractor neurons due to random spiking noise.

In the recurrent phase, performance of a given network's memory can be assessed by cueing certain subsets of minicolumns. This is done like during training (with help of the b_{on} flag), only this time allowing the network's recurrent connections to impose post-synaptic potentials according to their learnt synaptic weights, potentially converging network activity towards a memorized attractor. Methods for evaluating network performance will be further discussed in subsequent sections.

3.3 Model tuning & parameter scaling laws

The origins of the numeric values of parameters used in this model, as shown in Tables 3.1, 3.2, 3.3 and 3.4, are twofold: while most of them stem from physiological recordings or have been taken from similar models of previous papers (Chrysanthis et al., 2025, Tully, Hennig, and Lansner, 2014, Fiebig, Herman, and Lansner, 2020), certain parameter values had to be adjusted to account for differences in model size compared to other implementations. When performing those manual parameter tunings, there are two core conditions that should be met, specifically in the context of plastic spiking attractor neural networks:

1. Imposing synaptic weights of plausible amplitudes onto the network should result in the network displaying stable attractors in a physiological spiking regime.
2. Having synaptic plasticity, the network should be able to manually learn those same weight values when correctly trained.

Here, plausible weight amplitudes refers to values that can be expected to be learnt by the BCPNN rule through positive or negative spiking correlation (see Section 3.2.3). The question of what spiking frequencies are considered to be in a "physiological regime" can vary quite substantially across implementations (as well as areas of the brain), but generally speaking, firing rates of below 100 Hz are acceptable (more on attractor performance evaluation in Section 3.5). Naturally, this tuning process can be quite complex, as there possibly exist non-linear relationships between the constellation of two or more parameters and their resulting effect on network activity. However, remembering core principles about spatial and temporal summation of post-synaptic potentials (PSPs) can help guide the parameter tuning process. Physiological recordings have been made (Thomson et al., 2002) that analyze, for different kinds of cortical cells, different synapse types as well as in different cortical layers, how strongly a single excitatory or inhibitory post-synaptic potential (EPSP/IPSP) de- or hyperpolarizes the membrane on average, and how long these transient changes in membrane potential last. Computational model implementations of biological neural networks have long used these recordings as a reference for their tuning of synaptic conductances and current gains, and while exact curve fits are not always straightforward to obtain *in silico*, these recordings provide a good guideline for ensuring physiological spiking regimes.

Tuning single PSPs based on cell recordings naturally requires to be at the same time aware of connection probabilities inside the modeled network. Given the neural principle that it requires spatial summation of a certain number of EPSPs to cause the membrane potential to exceed the spiking threshold, the same physiological parameter values taken from physiological recordings won't be sufficient in their de- and hyperpolarizations if less than the adequate amount of incoming and outgoing synaptic connections is present at each neuron in the simulated network. Regarding long-range inter-hypercolumn connections, which contribute the largest number of synapses in this model, previous papers with comparable models (Chrysanthidis et al., 2025, Lundqvist, Herman, and Lansner, 2011) provide a good rule of thumb: According to those papers, each neuron inside an active attractor should receive roughly one hundred glutamatergic synapses from co-active neurons, i.e. neurons participating in the same attractor (locally speaking, i.e. within the simulated slice of neural tissue). Given that each memorized attractor would consist of one active minicolumn per hypercolumn, and that each pyramidal neuron receives around 10 synapses from pyramidal neurons within the same minicolumn (which are by definition co-active), one can define the formula for connection probability $p_c^{InterHC}$ as follows:

$$p_c^{InterMC} \times (N_{HC} - 1) \times N_{PYR} = 90 \iff p_c^{InterHC} = \frac{90}{(N_{HC} - 1) \times N_{PYR}} \approx 38\% \quad (3.16)$$

For the simulated network of the architecture and size as shown in Table 3.1, all parameter values declared in the previously mentioned tables hold true. However, as one of the main questions of this thesis is how to maintain attractor stability while

reducing the amount of simulated neurons per minicolumn, certain parameters will have to be modified in subsequent sections, following the parameter tuning guidelines declared in this section. Below is a list recapitulating each affected parameter, explaining why and how it should be tweaked to maintain global network dynamics as similar as possible to those of the control case. It should be noted that the declarations made in this list are merely hypotheses, meaning that it is possible that some of these proposed parameter changes will not have the expected effect on / necessity for the maintenance of model performance. Conclusions will be made in Section 4.3 regarding which of these parameter changes are most suitable for maintaining model performance in smaller networks.

1. PYR and BA count per minicolumn: As N_{PYR} is the manipulated variable, N_{BA} should be varied alongside it, to keep the ratio of basket to pyramidal cells roughly constant across simulations.
2. Pyramidal connection probabilities $p_c^{InterHC}$ and $p_c^{IntraMC}$: As explained in this section as well as Section 3.1, to follow the guidelines for the number of incoming positive synapses onto a given pyramidal cell (≈ 100), one can calculate connection probabilities in function of the network size. Naturally when this network size varies, connection probabilities will have to vary accordingly.
3. Pyramidal-basket synaptic conductance G_{PB} : To account for less and less neurons driving basket cells through EPSPs as the N_{PYR} decreases, the synaptic gain modulating each post-synaptic potential at PYR-BA synapses should be scaled up. This way, the effect of the WTA dynamics imposed by the basket cells can be kept constant. The scaling factor should be proportional to the ratio of original N_{PYR} to new N_{PYR} .
4. Recurrent pyramidal-pyramidal synaptic gain g_{gain}^{syn} : If at a certain point of N_{PYR} reduction the number of neurons still present in the network is so small that Equation 3.16 yields probability values above 1, then the connection probability will be capped at 1, effectively turning the network into a fully connected one (at least for inter-HC connections). To account for the impossibility to add more synapses to the network, and thus maintain a constant amount of total current passing through each pyramidal neuron, the synaptic gain for those synapses thus needs to be increased. Here, the factor of increase will be equal to the result of Equation 3.16, to keep the product of "neuron connections times current per neuron" constant. This tweaking step however only matters for very small values of N_{PYR} .

3.4 Model Implementation

The computational neural model developed in this thesis is a reiterated and adapted version of previously published spiking neural networks which combine the BCPNN learning rule with a biologically plausible columnar network organization as found in the mammalian neocortex. Direct inspirations for the model at hand include the models from the following papers: Fiebig, Herman, and Lansner (2020), Chrysanthidis et al. (2025) as well as Fiebig and Lansner (2017).

However, one notable difference to previous models is the neural simulator used in this thesis: *brian2* (Stimberg, Brette, and Goodman, 2019). The previously mentioned of comparable nature used as implementation references were created using NEST (Gewaltig and Diesmann, 2007), an advanced HPC neural simulator. As this

model represents a pioneer computational implementation, certain notations and formulas had to be rewritten to best represent how the actual model code is declared inside `brian2`.

One important feature that `brian2` offers are customizable spike events, meaning that specific code can be written for a given group of neurons or synapses, defining 1) conditions that should trigger a spike event (in the case of neuron groups), and more importantly 2) instructions that will be executed at the level of a neuron or a synapse whenever a spike occurs. This makes the explicit use of both Dirac delta as well as Heaviside step functions inside the model code obsolete, as changes in synaptic or neuron parameters in function of momentary spike events can be rewritten only with differential equations using *S*-notation. In the author's opinion, defining the entire model only in function of momentary changes with the help of spike indicator variables makes for model equations that are more easily understood and interpreted in a biological context. Biological systems seldom rely on time indices to update their internal states, but rather follow kinetic laws mathematically best described using differential equations.

3.5 Pattern generation & performance evaluation

Two terms used throughout this thesis are those of *pattern* and *attractor*. While they might appear to describe the same thing, a distinction between the two should be made: In the scope of this thesis, attractor refers to a state of network activity where a certain set of minicolumns (one per hypercolumn) maintains a higher-than-baseline firing activity over a longer period of time, surviving the end of any external cueing stimulus, while displaying attractor basin properties (see Section 2.1).

A pattern on the other hand refers to a set of coordinates pointing to a number of minicolumns that, after training the network by simultaneously exposing those exact minicolumns to external input leading to increased firing activity, might learn a stable attractor encompassing said minicolumns. Mathematically, a given pattern Q can be expressed as:

$$Q = \{\{h, m\} | h \in [1, N_{HC}], m \in [1, N_{MC}]\} \quad (3.17)$$

Given the idea that inside each hypercolumn, each minicolumn represents one of several possible values that the hypercolumn can code for, and that thus no two minicolumns of the same hypercolumn should participate in the same attractor (a principle also enforced by lateral inhibition from basket cells, see Section 2.2), a constraint for all patterns used for training attractors is that for each pattern Q , all minicolumns should be part of distinct hypercolumns. At the same time, each pattern should span all hypercolumns, meaning that all hypercolumns should be represented in a given pattern exactly once.

$$\begin{aligned} \forall Q, \forall \{h_1, m_1\}, \forall \{h_2, m_2\} : (\{h_1, m_1\} \in Q \wedge \{h_2, m_2\} \in Q) \implies h_1 \neq h_2 \\ |Q| = N_{HC} \end{aligned} \quad (3.18)$$

A set of patterns is considered orthogonal if no minicolumns participate in more than one pattern. The simplest set of orthogonal patterns that can be defined for a given network size is one where the i -th pattern (i from 1 to N_{MC}) consists of all i -th minicolumn of each respective hypercolumn. Thus, the maximum number of orthogonal patterns that can be stored in a network is equal to N_{MC} . When wanting to

store more patterns than that in the network, naturally, pattern overlaps will have to be introduced. This means that while the principle of only one allowed minicolumn per hypercolumn in a given pattern still holds true, a given minicolumn is allowed to participate in more than one pattern of a given pattern set. In theory, this can make learning stable attractors harder, as there might now be possibly be competition between the activation of two or more attractors when the network is cued with a distorted or incomplete set of minicolumns.

When evaluating a network regarding the successful establishment of a stable attractor for a given pattern after training, two conditions can be declared which together deem the attractor convergence successful or not:

1. *Reconstruction Success*: The condition is met if for a certain time interval (here chosen to be t_{ISI} milliseconds) after the end of a given pattern cue, the average firing rates of all neurons inside minicolumns participating in the respective pattern are above a certain value. Here, this value is chosen manually as $> 20\text{Hz}$.
2. *Interference*: The condition is met if for the same time interval, at least one minicolumn not participating in said pattern fires at an elevated rate, here chosen to be $> 10\text{Hz}$.

The firing rates chosen for these two conditions might seem a bit arbitrarily chosen, and while the exact frequencies that can be expected from an attractor made up of attractor-overlapping minicolumns might vary on a case-by-case basis, it is necessary to keep these values constant once chosen, to be able to make statistical inferences and comparisons between conditions (see Chapter 4). For a given network and tested pattern, running a certain number of independent distortion trials allows the calculation of the *reconstruction success rate* as well as the *interference rate*, two measures that will be used for statistically comparing memory stability across varying network sizes and parameter values.

In the following, three kinds of cueing will be performed to assert the attractor stability for a given pattern, asserting the two conditions defined above:

1. *Full cue*: Refers to shortly cueing (for t_{cue} milliseconds) the entire (non-distorted) pattern
2. *Incomplete cue*: Refers to cueing only a certain subset of all in-pattern minicolumns
3. *Distorted cue*: Refers to cueing one minicolumn per hypercolumn where a certain number of minicolumns have been randomly resampled to other minicolumns, not participating in the current pattern

Chapter 4

Results

4.1 Simulations with orthogonal patterns

As a first step to ensure that the model is able to properly learn stable attractors, orthogonal patterns are generated and used to train the network.

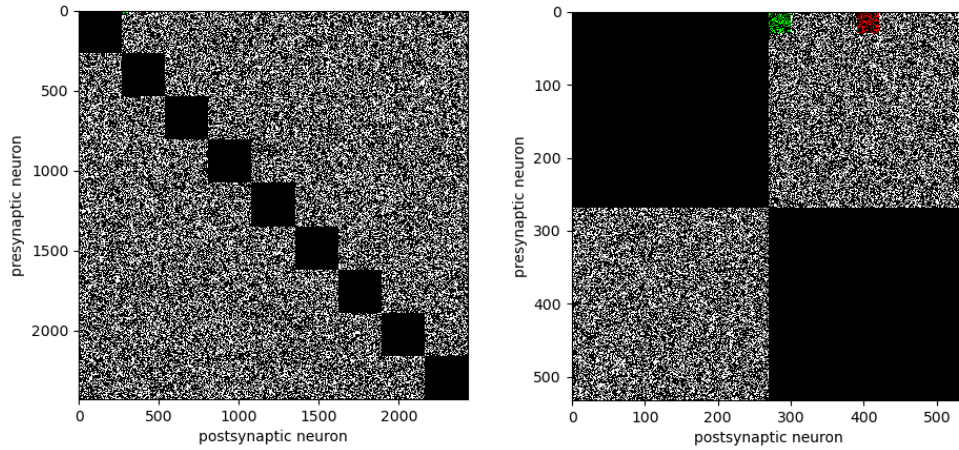


FIGURE 4.1: Inter-hypercolumn PYR-PYR connectivity in a network of size 9x9x30. **Left panel:** Connectivity across all neurons. **Right panel:** Connectivity between first two hypercolumns. White pixel = synapse present; black pixel = synapse absent. Green and red areas in right panel represent synapses whose weights are averaged to create the weight trajectories shown in Figure 4.2.

Figure 4.1 shows the connectivity of plastic inter-HC synapses for the network instantiated in this section and subsequently trained on orthogonal patterns. As stated in Methods, average connectivity across hypercolumns is homogeneous and follows the connection probability declared in Table 3.1. The spiking raster plot in Figure 4.2 shows the process of training all nine orthogonal patterns (here the patterns were presented to the network in randomized order).

The bottom panel of Figure 4.2 shows the trajectories over time of averaged weights of all synapses between minicolumns {1,1} and {2,1} (green line) as well as all synapses between minicolumns {1,1} and {2,5} (red line). The differences in weight trajectories can be explained by the fact that the first pair of minicolumns are part of the same trained pattern (last pattern trained, see top panel), while the second pair of minicolumns are in different patterns (minicolumn {2,5} is part of the second trained pattern). Due to the BCPNN learning rule (see Section 3.2.3), the first set of synapses will develop positive weights due to strong positive spiking correlations

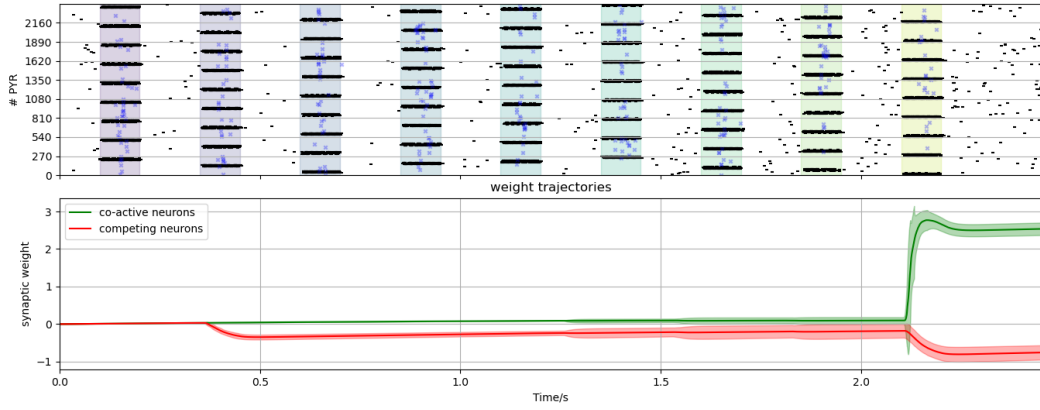


FIGURE 4.2: Orthogonal pattern training. **Top panel:** Raster plot of all pyramidal (black dots) and basket (blue crosses) cells over training time. Faint horizontal lines separate hypercolumns. Colored vertical bars indicate times of pattern stimulation where each color corresponds to a different pattern (trained in randomized order). **Bottom panel:** Two example weight trajectories for mean weights of minicolumn-to-minicolumn synapse groups (as seen in Panel 2 of Figure 4.1). Colored area around each of the two curves corresponds to respective standard deviation.

between all co-active minicolumns, while the second set of synaptic weights will become negative due to negative spiking correlations between minicolumns {1,1} and {2,5}.

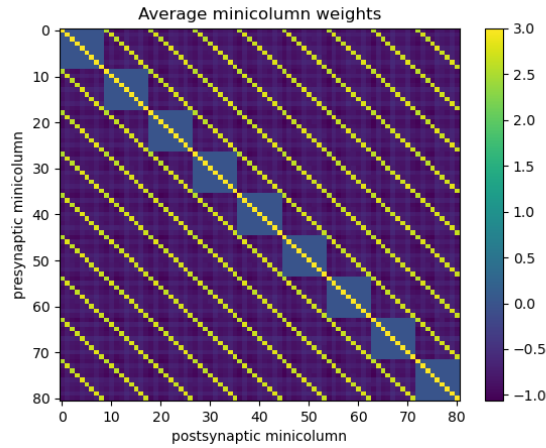


FIGURE 4.3: Weight matrix resulting from training orthogonal patterns. Numeric values correspond to the results of the BCPNN learning rule (logarithms of P-trace ratios) and are not yet multiplied with respective synaptic gain as in Equation 3.12.

Figure 4.3 shows the global weight matrix for all PYR-to-PYR synapses of the network, resulting from the orthogonal pattern training seen in Figure 4.2. For each pair of minicolumns, weights of all synapses are averaged. The yellow diagonals correspond to very strong weights corresponding to those minicolumn pairs that are within the same respective patterns, as well as the non-plastic intra-MC weights on the main diagonal. The dark purple diagonal stripes represent negative weights

learned between all minicolumn pairs participating in respectively different patterns. Lastly the cyan squares along the main diagonal show the absence of any synapses for any pair of distinct minicolumns that are inside the same hypercolumn (see Section 3.1).

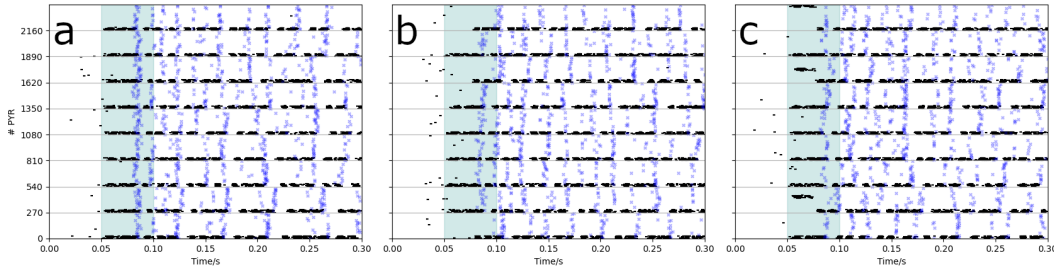


FIGURE 4.4: Raster plots of example memory tests for network trained on orthogonal patterns. **a:** Simple cueing test; The colored horizontal bar corresponds to the cue, comprised of a full pattern. **b:** Completion test; The cue consists of only six out of nine minicolumns. **c:** Distortion test; A distorted pattern is cued, where three minicolumns have been randomly resampled across their respective hypercolumns.

Now, with a network having those weights shown in Figure 4.4, memory tests as defined in Section 3.5 can be performed. Figure 4.4 shows example raster plots for three kinds of tests on the first pattern (cueing the entire pattern, cueing an incomplete pattern and cueing a distorted pattern). It is apparent that the attractor corresponding to this pattern has been properly learnt and is structurally as well as temporally stable. For all three cases, long after the respective cue (light blue vertical stripe) has stopped, the correct minicolumns maintain a state of increased spiking activity, while all other minicolumns remain silent. It can also be seen that the basket cell population across the network engages in rather synchronous bursting activity, driven by the very strong inter-HC synaptic connections between in-attractor synapses. It can be seen how each basket cell burst temporarily silences all pyramidal activity in its respective hypercolumn, which however is not sufficient to turn off the attractor, most likely thanks to the variations in synaptic conduction delay implemented as declared in Table 3.1 (if all basket cells were to spike at the exact same time, the attractor would be silenced fully, without any one minicolumn still spiking to turn the rest back on). One last thing to note in this figure is that for the completion as well as distortion tests (panels b and c), the respective non-cued in-attractor minicolumns already start activating even before the cue has finished. In the case of distortion, the "wrong" (but cued) minicolumns even completely shut off before the end of the cue, likely due to both the strong inhibition from other active minicolumns, as well as the intra-MC competition implemented by the basket cells.

4.2 Simulations with 20 random patterns

Moving on from orthogonal patterns, minicolumn overlaps will be introduced to allow for the storage of more than 9 patterns. In the following, two independent simulation runs will be performed, respectively on two previously blank networks, with respectively two sets of randomly generated patterns (20 total patterns each) called pattern set A and B.

Both simulation runs will consist of the following steps:

1. Generation of 20 random patterns (see Section 3.5). The number 20 is fixed to allow for comparisons in network performance across pattern sets, as random pattern overlaps could influence average network performance for a given pattern set.
2. *Training phase*: Like in Section 4.1, expose each of the 20 patterns to the network one by one, by turning on external excitatory input of all in-pattern minicolumns. At the end of the simulation, save all resulting synaptic Z- and P-traces into a data file for later model reconstruction.
3. *Recall phase*: For N_{iter} iterations, perform the following steps:
 - (a) Instantiate a new network with the saved synaptic trace values from step 2.
 - (b) Randomly select a pattern from the current pattern list to perform a distortion test (see Section 3.5).
 - (c) Save the distortion test result as well as the coordinates of both the original and the corresponding distorted pattern to a .csv file for later analysis.

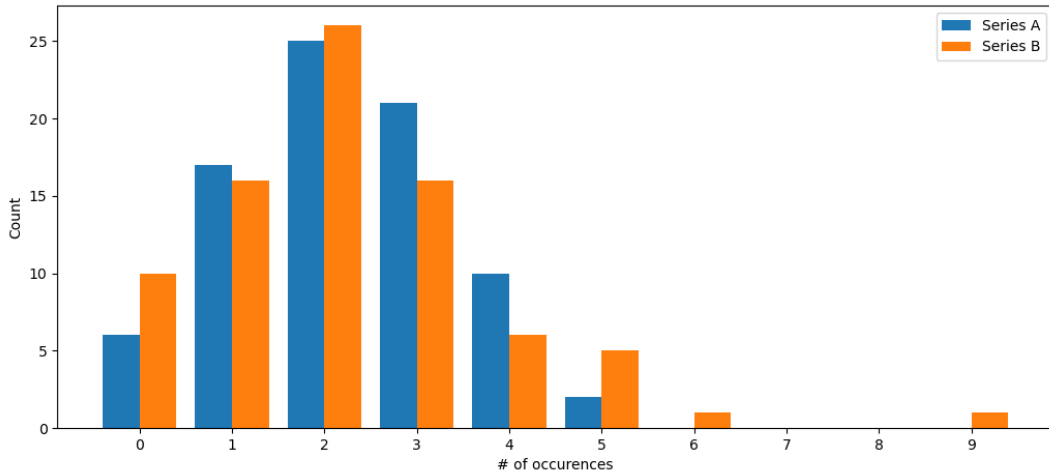


FIGURE 4.5: Histograms of minicolumn occurrences for two sets of 20 randomly generated patterns.

Figure 4.5 recapitulates how often each of the networks $9 \times 9 = 81$ minicolumns occurs in each of the two sets of 20 non-orthogonal patterns. For both pattern sets, most minicolumns appear between one and three times, while some appear more than that and some do not appear at all. For pattern set B, the maximum number of times that a minicolumn appears in the set is nine (compared to five for set A), meaning that one minicolumn appears in almost half of all patterns subsequently stored in the network.

Figure 4.6 shows the synaptic weights that result from respectively training a network either on pattern set A or pattern set B. A striking difference between these new weight matrices to those of the previous section is the absence of clear diagonal lines (apart from the main diagonal representing nonplastic intra-MC weights), due to the absence of perfectly orthogonal patterns in either pattern set. It appears that in addition, the mean of positive weights is smaller compared to the weight matrix in

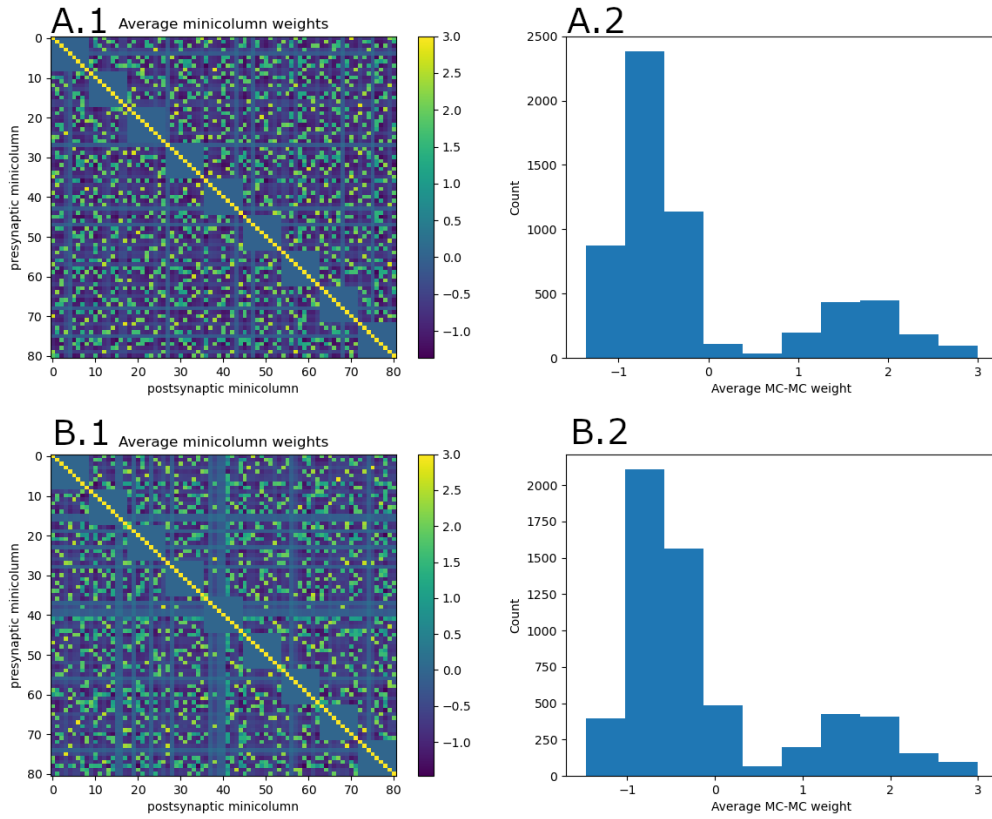


FIGURE 4.6: Weights after two independent training runs with different sets of randomly generated patterns. **A.1 & B.1:** BCPNN weight matrices averaged for each MC-to-MC synapse group. **A.2 & B.2:** Histograms of total occurrences of weight values, averaged for each MC-to-MC synapse group.

Figure 4.3 (judging by the amount of yellow versus green pixels in either weight matrix). This makes sense considering that in the case of non-overlapping patterns, it is possible that two minicolumns will appear in a given pattern together, increasing their inter-MC weights during training of that pattern, followed by a pattern where only one of those same minicolumns appears, subsequently lowering their weight again slightly (as can be seen in the example weight plot of Figure 4.2). On average, while still being positive, weights between minicolumns that sometimes (but not always) appear in the same pattern will thus be lower than weights between minicolumns that are exclusively coactive.

After performing 200 independent distortion trials for each of the two networks (simulation setup as mentioned above, with $N_{iter} = 200$), average reconstruction success as well as interference rates can be computed. Figure 4.7 shows these two rates respectively for both simulation runs. Success rates for both batches of trials are very close to 100 per cent (see Figure A.1), with no strongly statistically significant difference in reconstruction success ($p > 0.01$) or interference rates. For this section, a conclusion can be made that the model is generally able to successfully learn and (almost perfectly) recall at least 20 overlapping patterns. These results will provide a control condition for the next section, where the network size will gradually be reduced.

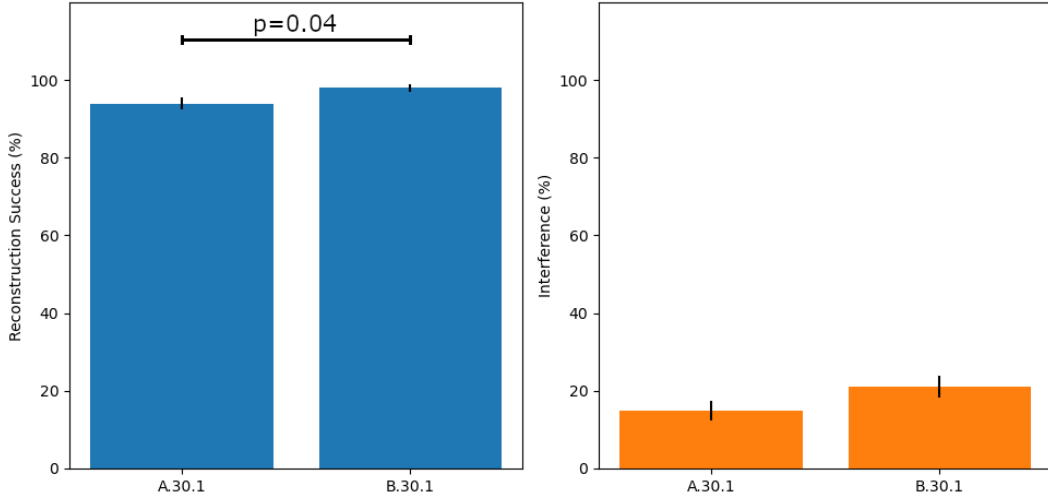


FIGURE 4.7: Average reconstruction (left panel) as well as interference (right panel) rates for simulation runs with pattern sets A and B. Black lines for each bar represent respective standard deviations of underlying Binomial distributions and are indicated in supplementary tables 1 and 2 (Figure A.1).

4.3 Reducing N_{PYR}

For the last part of this thesis, experiments will be run to assert network memory performance when reducing the amount of pyramidal neurons per minicolumn N_{PYR} . Specifically, for each simulation run, or batch, with a given network size and set of studied parameters, the overall reconstruction success rate interference rates will be compared to the corresponding rates of the original size network from Section 4.2. Thus, in order to be able to directly compare these rates, the same two sets of non-overlapping patterns (set A and set B) will be used for network training.

In subsequent batches (each batch following the simulation structure defined in Section 4.2), N_{PYR} will be reduced to 15, 10 and finally 5. In addition to a varied N_{PYR} , each batch will also feature a distinct combination of varied parameter values among the parameters identified in Section 3.3, and be run twice, for two networks respectively trained on pattern sets A and B. The full table of batches specifying all varied parameters as well as final reconstruction success and interference rates can be found in the supplementary material (Figure A.1). The notation of batches follows the structure of "X.Y", where X is the current N_{PYR} and Y enumerates all batches with that given N_{PYR} . For each of the two pattern sets, there are four batches for 15 PYR neurons per MC, thus ranging from 15.1 to 15.4, three for 10 PYR neurons (10.1 to 10.3) and two for 5 PYR neurons (5.1 and 5.2). Not all possible combinations of network parameters among those selected for variations were tested in simulations, but enough variations exist across simulations to infer causal relationships between each given parameter and its respective effect on memory performance.

Figures 4.8 and 4.9 recapitulate and compare the differences in average reconstruction and interference rates for all batches for pattern set A and B respectively. All batches corresponding to a given value of varied N_{PYR} (15, 10 and 5) are statistically compared to each other as well as the control condition of $N_{PYR} = 30$.

The first thing to note is that between pattern sets A and B, the statistically significant differences in reconstruction success rates within each panel are exactly the same, while this is not true for differences in interference rates. This indicates that

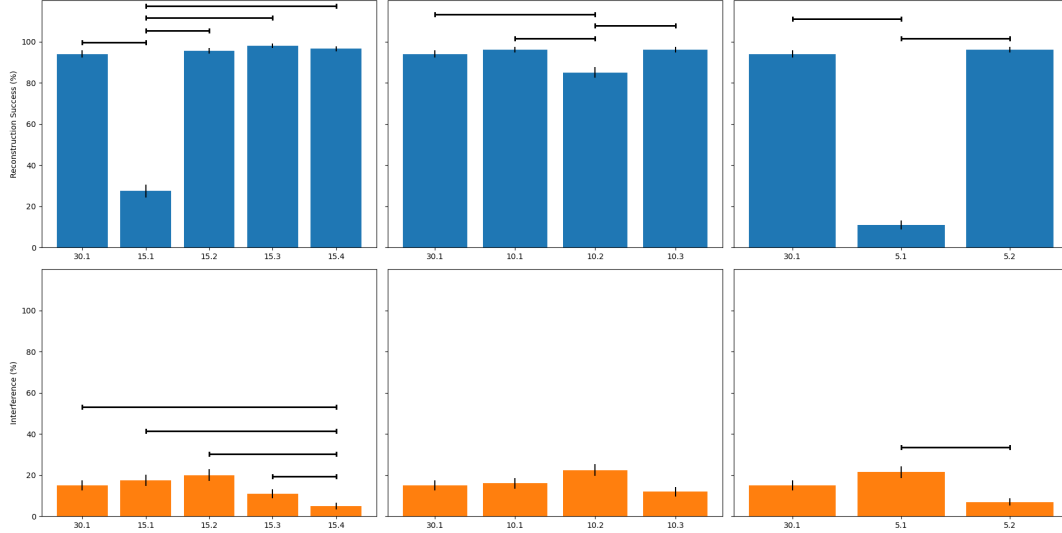


FIGURE 4.8: Bar plot grouping together batch reconstruction success rates (top row) and interference rates (bottom row) as well as respective significance statistics for batches $N_{PYR} = 15$ (left column), $N_{PYR} = 10$ (center column) and $N_{PYR} = 5$ (right column) for all simulation runs with pattern set A. Each panel compares respective batch rates to control batch of $N_{PYR} = 30$. Horizontal bars indicate statistically significant difference in rate ($p < 0.01$), see supplementary tables 3 to 6 (Figs. A.2 & A.3) for full statistical comparisons.

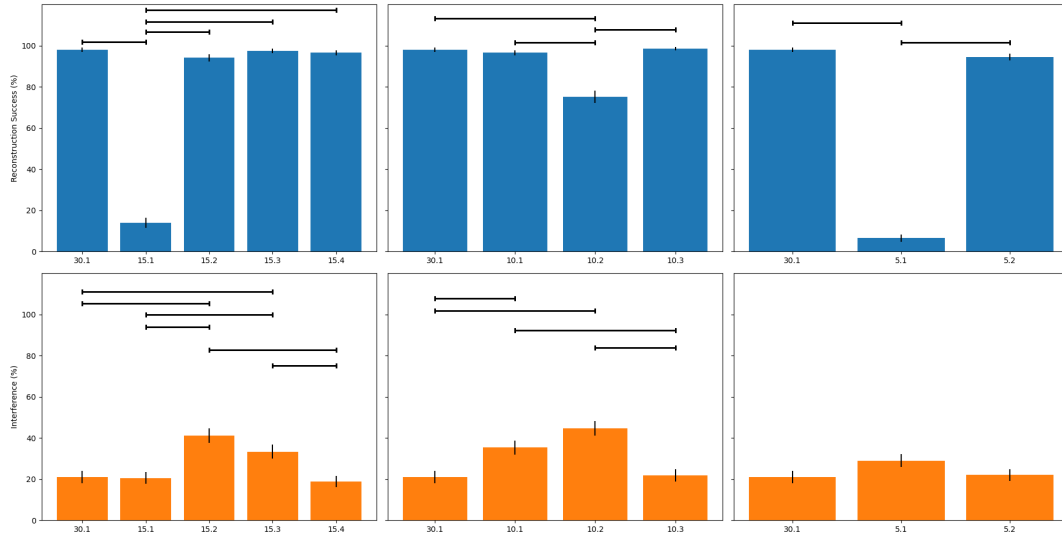


FIGURE 4.9: Same bar plot as Figure 4.8, but for all simulation runs with pattern set B.

differences in reconstruction success rates among batches are more likely to be solely due to parameter differences, rather than being due to random differences in pattern constellations. The latter does appear to have a larger influence on interference rates, which makes sense considering that interferences of non-attractor minicolumns actively spiking depend on whether or not one or more minicolumns of the currently active attractor have strong synaptic connections to those non-attractor minicolumns, something that depends on the random initialization of the current pattern set.

When scaling down from 30 to 15 PYR neurons per minicolumn without changing any other network parameters, network performance drastically decreases to around 30% reconstruction success. Scaling $p_c^{InterHC}$ up according to the guideline defined in Section 3.3 reverts this effect. For this network size, scaling up the other parameters (intra-MC connectivity and PYR-BA synaptic gain) will not have a significant effect on reconstruction success rate, at least as long as inter-HC connectivity has already been scaled up. Scaling up PYR-BA synaptic gain does however have a significant effect on reducing interference rate in this case. When reducing N_{PYR} further down to 10, assuming that inter-HC connectivity has already been scaled accordingly, there is only significant difference in reconstruction success rates between batches of varying intra-MC connectivity, not between batches of varying PYR-BA synaptic gain. Scaling the latter does again (at least in batches with pattern set B) have a statistically significant effect on minimizing the average interference rate. Lastly, in the last set of batches, where N_{PYR} has been scaled down to 5 and all other previously mentioned parameters have been scaled according to the declared guidelines, scaling up synaptic gain g_{gain}^{syn} is necessary to maintain reconstruction performance. For pattern set A, scaling the synaptic gain also has a significant effect on reducing the amount of interference.

While not all possible parameter scaling combinations have been explored in separate simulation runs, making it impossible to perfectly infer the necessity as well as sufficiency of each scaling operation for model performance, a certain "hierarchy" of tested parameters can be declared, determining the order of attention that should be given to each parameter when scaling down the size of a given simulated network: First, inter-hypercolumn connectivity should be scaled according to Equation 3.16. Second, if the result of this equation yields impossible connection probabilities of above 100%, synaptic gain of PYR-to-PYR synapses needs to be scaled by this result, to account for a lack of "space" in the network for synapses to be instantiated. Third, intra-minicolumn connectivity should be scaled along with the inter-HC connectivity. Lastly, PYR-to-BA synaptic gain should be scaled by a factor of $\frac{30}{N_{PYR}}$, although this scaling will have less of an effect on reconstruction success rate, and more so on minimizing interference from non-attractor minicolumns, due to the soft WTA dynamics imposed by the basket cells.

It can be generally said that when performing all of these scaling operations, the resulting network performance stays constant, without any statistically significant differences (except for one case where scaling actually significantly reduced interference rate compared to the control case - pattern set A, batch 15.4). This suggests that the set of selected parameters are sufficient for keeping network performance constant across minicolumn sizes even smaller than those tested in this thesis, assuming they are scaled correctly.

Chapter 5

Discussion

The results presented above provide a clear answer to the research questions and working hypotheses defined in Section 1.2: Spiking neural networks of columnar architecture simulating individual neurons instead of minicolumn neuron populations are able to successfully learn and recall associative memories through Hebbian learning, while minimal set of network parameter scaling operations can be performed to maintain attractor memory performance even when drastically reducing the amount of neurons simulated for each minicolumn. There are, however, points that should be kept in mind while considering these results. First, while the tuning of the parameters proposed works as predicted, there are many more network parameters that were kept constant. It is possible that there exist other sets of parameter tunings which would have had similar effects on memory performance. Thus, it cannot be concluded without a reasonable doubt that these proposed tuning operations are truly necessary for maintaining memory in a shrinking network. Secondly, it remains possible that the original sets of parameter values used in Sections 4.1 and 4.2, and considered the control condition in Section 4.3, were never the most optimal parameters to begin with. Finding stable regimes of spiking activity in this kind of spiking neural network is a tedious task, and there exist many hardly predictable relationships between any two or more parameters regarding network performance. While the network state considered here as the control case had reasonably good memory performance, minicolumn interferences were still quite high: Around 20% of all trials had some sort of interference, a number which could have possibly reduced to much less. And it is possible that in the case of a more optimal control network, memory performance effects of the performed scaling operations would have been significantly different.

Another question to keep in mind, and really the question that motivated the scaling experiments performed in this thesis, is that of simulation run time. The original premise stated that bigger networks are more stable in their physiological behavior and thus more suited for neuromorphic computations interesting for research and application, yet at the same time, running bigger networks takes longer. Thus, the interest to scale down network size (and thus computation time) while maintaining network performance is of a central interest. However, this premise does not provide insight into the question whether it is neuron count, or rather synapse count, that determines the amount of time it takes to run a given computational neural network. If the latter is true, then the performed scaling operations do not actually reduce computation time in cases where connection probability scalings are the only parameters necessary to maintain network performance (for example when going from 30 to 15 pyramidal neurons per minicolumn in a network 9×9 minicolumns). In order to assess the real impact of the work presented in this thesis on computation time in large-scale networks, new studies will have to be performed that collect data and compute statistics on the average execution times for networks

of different sizes and parameter sets.

In light of the most recent economic growth observed in the AI industry, questions and concerns regarding sustainability should be raised. Large AI models built by private tech companies running on enormous data centers have become a non-negligible part of global energy consumption (Vries-Gao, 2026). And while one of the main research incentives in the field of neuromorphic computing is that of more energy-efficient models, this does not necessarily mean that breakthroughs in AI model energy and cost efficiency would actually lead to less global energy consumption in the long run, as private companies would naturally try to maximize their production output and thus energy consumption either way. In other words, if the same amount of AI computation costs less energy (and thus money) to produce, why not produce more computation for the same price, and thus get more returns? In modern times, machine learning researchers should be aware of the very real effects that newer and better models and algorithms can have outside the narrow scope of academia. But then of course, a counter-argument can be made that a better understanding of the brain is always good for society as a whole, as it allows for better medical procedures to be developed, including surgical, pharmaceutical as well as therapeutical methods.

Appendix A

Supplementary Figures

Suppl. Table 1: Simulations with Pattern Set A								
Batch	N_iter	neurons	p_c	p_c_MC	G_PB	syn_gain	Recon (%)	Interf (%)
30.1	200	30+4	38.00%	25.00%	x1	x1	94±1.68	15±2.52
15.1	200	15+2	38.00%	25.00%	x1	x1	27.5±3.16	17.5±2.69
15.2	202	15+2	75.00%	25.00%	x1	x1	95.5±1.47	20±2.83
15.3	201	15+2	75.00%	50.00%	x1	x1	98±0.99	11±2.21
15.4	201	15+2	75.00%	50.00%	x2	x1	96.5±1.3	5±1.54
10.1	201	10+2	100.00%	75.00%	x1	x1	96±1.39	16±2.59
10.2	201	10+2	100.00%	25.00%	x3	x1	85±2.52	22.5±2.95
10.3	201	10+2	100.00%	75.00%	x3	x1	96±1.39	12±2.3
5.1	200	5+2	100.00%	100.00%	x6	x1	11±2.21	21.5±2.9
5.2	200	5+2	100.00%	100.00%	x6	x2.2	96±1.39	7±1.8

Suppl. Table 2: Simulations with Pattern Set B								
Batch	N_iter	neurons	p_c	p_c_MC	G_PB	syn_gain	Recon (%)	Interf (%)
30.1	200	30+4	38.00%	25.00%	x1	x1	98±0.99	21.0±2.88
15.1	200	15+2	38.00%	25.00%	x1	x1	14±2.45	20.5±2.85
15.2	202	15+2	75.00%	25.00%	x1	x1	94.06±1.66	40.1±3.45
15.3	201	15+2	75.00%	50.00%	x1	x1	97.51±1.1	33.83±3.34
15.4	201	15+2	75.00%	50.00%	x2	x1	96.52±1.29	18.91±2.76
10.1	201	10+2	100.00%	75.00%	x1	x1	96.52±1.30	35.32±3.37
10.2	201	10+2	100.00%	25.00%	x3	x1	75.12±3.05	44.78±3.51
10.3	201	10+2	100.00%	75.00%	x3	x1	98.51±0.86	21.89±2.92
5.1	200	5+2	100.00%	100.00%	x6	x1	6.5±1.74	29.5±3.22
5.2	200	5+2	100.00%	100.00%	x6	x2.2	94.5±1.61	22.0±2.93

FIGURE A.1: Chosen network parameters as well as average reconstruction success and interference rates for all batches with pattern sets A and B (respectively first and second table).

Suppl. Table 3: p-Value of Reconstruction Success: Pattern Set A									
15.1	15.2	15.3	15.4	10.1	10.2	10.3	5.1	5.2	Batch
8.44E-56	0.5	0.04	0.24	0.36	3.00E-03	0.36	1.94E-103	0.36	30.1
	7.29E-60	1.56E-67	8.72E-63	2.66E-61	2.91E-37	2.66E-61	2.46E-05	2.66E-61	15.1
		0.16	0.61	0.8	3.76E-04	0.8	3.65E-111	0.8	15.2
			0.36	0.24	2.45E-06	0.24	1.61E-126	0.24	15.3
				0.79	6.43E-05	0.79	6.95E-117	0.79	15.4
					1.61E-04	1	5.82E-114	1	10.1
						1.61E-04	1.03E-70	1.61E-04	10.2
							5.82E-114	1	10.3
								5.82E-114	5.1

Suppl. Table 4: p-Value of Reconstruction Success: Pattern Set B									
15.1	15.2	15.3	15.4	10.1	10.2	10.3	5.1	5.2	Batch
8.16E-111	0.04	0.74	0.36	0.36	5.62E-12	0.69	6.90E-160	0.06	30.1
	4.03E-92	1.88E-108	3.43E-103	3.43E-103	4.75E-43	3.44E-114	0.01	1.92E-93	15.1
		0.08	0.25	0.25	9.03E-08	0.02	1.71E-128	0.85	15.2
			0.56	0.56	2.17E-11	0.48	1.63E-155	0.12	15.3
				1	3.37E-10	0.2	1.69E-146	0.33	15.4
					3.37E-10	0.2	1.69E-146	0.33	10.1
						1.02E-12	8.18E-60	4.09E-08	10.2
							8.37E-166	0.03	10.3
								8.74E-131	5.1

FIGURE A.2: Statistics of reconstruction success rate comparisons between batches. Dark green cells indicate strongly significant results ($p < 0.01$) while faint green indicate cases where $p < 0.05$

Suppl. Table 5: p-Value of Interference: Pattern Set A									
15.1	15.2	15.3	15.4	10.1	10.2	10.3	5.1	5.2	Batch
0.45	0.19	0.24	8.19E-04	0.78	0.05	0.38	0.09	0.01	30.1
	0.52	0.06	6.81E-05	0.69	0.21	0.12	0.31	1.31E-03	15.1
		0.01	4.62E-06	0.3	0.54	0.03	0.71	1.30E-04	15.2
			0.03	0.14	2.01E-03	0.75	4.35E-03	0.16	15.3
				3.11E-04	2.60E-07	0.01	8.40E-07	0.4	15.4
					0.1	0.25	0.16	4.71E-03	10.1
						0.01	0.81	1.03E-05	10.2
							0.01	0.09	10.3
								2.91E-05	5.1

Suppl. Table 6: p-Value of Interference: Pattern Set B									
15.1	15.2	15.3	15.4	10.1	10.2	10.3	5.1	5.2	Batch
0.90	1.14E-05	5.43E-03	0.60	1.38E-03	2.84E-07	0.83	0.06	0.81	30.1
	6.42E-06	3.70E-03	0.69	8.98E-04	1.47E-07	0.73	0.05	0.71	15.1
		0.11	8.83E-07	0.23	0.46	2.91E-05	0.01	3.36E-05	15.2
			9.50E-04	0.68	0.02	0.01	0.35	0.01	15.3
				1.97E-04	1.50E-08	0.46	0.02	0.44	15.4
					0.05	2.82E-03	0.18	3.11E-03	10.1
						8.41E-07	1.02E-03	9.98E-07	10.2
							0.10	0.98	10.3
								0.11	5.1

FIGURE A.3: Statistics of interference rate comparisons between batches. Dark green cells indicate strongly significant results ($p < 0.01$) while faint green indicate cases where $p < 0.05$

Suppl. Table 7: p-Values for Batch-by-Batch Comparisons		
Batch	p-Value Reconstruction	p-Value Interference
30.1	0.04	0.12
15.1	8.32E-04	0.45
15.2	0.52	3.57E-06
15.3	0.74	4.63E-08
15.4	0.99	1.52E-05
10.1	0.79	7.82E-06
10.2	0.01	1.82E-06
10.3	0.12	0.01
5.1	0.11	0.08
5.2	0.48	1.74E-05

FIGURE A.4: Statistics of reconstruction success and interference rate comparisons between control conditions 30.1 for both pattern sets (A and B). Dark green cells indicate strongly significant results ($p < 0.01$) while faint green indicate cases where $p < 0.05$)

Appendix B

Internship Pre-Registration

Master Thesis Pre-Registration (January 2026)

Bujanowski, Ferdinand | Herman, Pawel

Computational Brain Science Lab, KTH Royal Institute of Technology, Stockholm, Sweden

Scientific Background and Context

Artificial neural networks are the backbone of many modern-day artificial intelligence algorithms and applications. The interest in making these ANNs more biologically plausible and thus more brain-like is two-fold: On one hand, given the vast learning capacities of the brain combined with its small energy requirements, adapting brain-like computing paradigms into ANNs could make them more energy-efficient and capable of handling real-time data. On the other hand, neuromorphic ANNs allow for the formulation and testing of hypotheses regarding the brain's organization, on various temporal and spatial scales, that would be impossible to verify *in vivo* (Ravichandran, Lansner, & Herman, 2025). In contrast to the comparably unbiological deep neural network (DNN) used on a large scale in modern machine learning applications, artificial recurrent spiking neural networks (SNNs) offer a biologically more plausible alternative in the sense that they model how biological neurons communicate with one another - via *spiking*, i.e. the emission of action potentials (Maass, 1997). A benchmark cognitive ability that the brain's spiking neural networks are capable of is the learning and maintaining of so-called **associative memory** (Hebb, 2005; Lansner, 2003), also called *attractor* (Hopfield, 1982) or *content-addressable memory*. This refers to memory that can be (re)activated by exposing it to distorted or partial input corresponding to the memorized object, or to input corresponding to an associated object (Lansner, Ravichandran, & Herman, 2025).

While they are promising models, getting SNNs to reliably learn sparse distributed internal representations from real-world data in an unsupervised manner, a task that the human brain is notably capable of and that DNNs succeed in, is an ongoing challenge. In a recent computational modeling study by Ravichandran, Lansner, and Herman (2024), a novel SNN was developed on the basis of a non-spiking rate-based model with brain-like architecture and local (Hebbian) learning rules (Lansner et al., 2025; Lansner & Ekeberg, 1989). Specifically, brain-like architecture refers to the organization of the network into functional units of hyper- and minicolumns, replicating the modular, columnar architecture of the mammalian neocortex (Mountcastle, 1997). Minicolumns, the smallest functional units of neural computation, were hereby modeled as Poisson spike generators with sparse firing (Buesing et al., 2011). This new SNN managed to learn distributed linearly separable representations using the MNIST dataset (Lecun et al., 1998) as learning input, and performed comparably to its non-spiking counterpart on common associative memory metrics such as pattern completion, perceptual rivalry as well as distortion resistance, using visually corrupted MNIST examples for evaluation.

Research Question(s) and Working Hypotheses

A logical next step towards more brain-like SNNs capable of sparse representation learning and associative memory is to introduce more biologically plausible sources of spiking to the aforementioned model. Minicolumns in the mammalian neocortex consist of many (around 80 to 100) neurons, which are tightly interconnected with one another (Buxhoeveden & Casanova, 2002). Two commonly used models of single neurons' intracellular voltage change and resulting spiking over time are the Leaky Integrate and Fire (or LIF) model (Lapique, 1907) as well as the Hodgkin-Huxley model (Hodgkin & Huxley, 1952). Both models use sets of differential equations to describe the neuron's electrophysiology, and are thus well suited for computational simulation. Previous studies showed that global spiking statistics of recurrent SNNs with columnar architecture modeled with LIF neurons can follow a Poisson distribution (Lundqvist, Compte, & Lansner, 2010).

It will thus be the research objective of this internship to study a novel version of this latest SNN, only this time replacing the single-unit Poisson spike generator minicolumns by minicolumns each grouping together several LIF neurons. It will not be studied whether this SNN, like its predecessor, is able to *learn* sparse distributed internal representations in an unsupervised manner using the same task and dataset, but rather whether an LIF-SNN having the same synaptic weights as well as connectivity patterns (taken directly from the Poisson-SNN) still displays associative memory of the same quality, using the same metrics of pattern completion, perceptual rivalry and distortion resistance for comparison. To conclude, the working hypothesis tested in this internship will be the following: *the SNN with LIF-neuron minicolumns, integrated within the same brain-like network architecture and initialized with pre-trained inter- and intra-column connection weights, achieves the same performance as the sparsely spiking Poissonian neuron network in terms of associative memory functionality.*

Another crucial question that will be explored in the internship is how many LIF neurons per minicolumn are needed for the model to successfully display the same properties of associative memory as its Poissonian predecessor. Simulating neurons individually instead of as an ensemble comes with a substantial additional computational cost. Naturally, the question of what the minimum number of LIF neurons per minicolumn necessary to ensure fully functional associative memory in the network arises. Thus, a second hypothesis that will be tested in this internship is the following: *when scaling down the number of neurons per minicolumn, the associative memory functionality of the network with decrease. This decrease will not follow a purely linear, but rather a logarithmic or logistic trend.*

FIGURE B.1: Pre-registration page 1

Planned Methodology

The LIF spiking neural network described above will be modeled using the Python library `brian2` (Stimberg, Brette, & Goodman, 2019). This library offers a lightweight and easy to use interface for simulating groups of neurons described on an individual scale using systems of differential equations, and is thus perfectly suited for the simulation of spiking neural networks of various sizes. All simulations will be run on the KTH supercomputer that is available to the Computational Brain Science Lab. In a first step, the precursor Poissonian SNN, which was developed in C++ and is available to the lab, will be replicated inside `brian2` in terms of its layer architecture as well as inter- and intra-hypercolumn connectivity and synaptic weights, the latter of which supposedly enabling transfer of the already learnt associative memory onto the new network. Then, each minicolumn will be populated with n LIF neuron units, the intra-minicolumn synapses of which will be initialized to be very strong, to ensure highly correlated spiking among minicolumn neurons. *A priori*, n will initially be in physiological range, meaning $n \in [80, 100]$, if the computational cost of simulation is not too high. Here, the relative number of excitatory and inhibitory neurons will be modeled after percentages commonly found in the mammalian neocortex (Sukenik et al., 2021). Once the LIF-SNN is fully established, the network's performance is measured on three distinct associative memory tasks - pattern completion, perceptual rivalry and distortion resistance - using the same modified MNIST datasets also used to evaluate the predecessor Poissonian SNN. Finally, in order to study the effect of minicolumn size on network performance, the amount of LIF neurons per minicolumn n will gradually be decreased while the same network performance measures will be made each time.

Expected Observables and Data

In order to evaluate the LIF-SNN on the functionality of its associative memory, as explained above, tests of pattern completion, perceptual rivalry and distortion resistance will be performed, just like it was done in the original study on the Poisson-SNN. For that, the authors created three variants of the MNIST dataset, in which the original images of handwritten digits are corrupted to varying degrees, corresponding to varying reconstruction difficulties. For a given task and image, it will be evaluated how similar the z-traces of the *input reconstruction* layer of the network are to the binary pixel data of the respective original uncorrupted MNIST image. For a given task and degree of difficulty, mean performance will be calculated by averaging over respective trials, and for each respective task the resulting difficulty-performance curve will be plotted.

When gradually scaling down the network size to test the second hypothesis, two observables will be gathered for each respective model version and image reconstruction task: 1) the maximum mean model performance (*a priori* corresponding to least reconstruction difficulty), and 2) the steepness of model performance decline resulting from increasing difficulty.

For a given reconstruction task, these observables will be plotted as a function of minicolumn size. In addition, a ratio of simulation computational cost (measured for example in time required for running a single model simulation) over mean total model performance is calculated and also plotted in function of minicolumn size, so as to find a potential optimum of the computation cost - performance tradeoff, which can then be utilized in future implementations of the LIF-SNN.

References

- Buesing, L., Bill, J., Nessler, B., & Maass, W. (2011, November). Neural Dynamics as Sampling: A Model for Stochastic Computation in Recurrent Networks of Spiking Neurons. *PLoS Computational Biology*, 7(11), e1002211. doi: 10.1371/journal.pcbi.1002211
- Buxhoeveden, D. P., & Casanova, M. F. (2002). The minicolumn hypothesis in neuroscience. , 125(5), 935–951. doi: 10.1093/brain/awf110
- Hebb, D. (2005). *The organization of behavior* (??th ed.). Psychology Press. doi: 10.4324/9781410612403
- Hodgkin, A. L., & Huxley, A. F. (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. , 117(4), 500–544. doi: 10.1113/jphysiol.1952.sp004764
- Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. , 79(8), 2554–2558. doi: 10.1073/pnas.79.8.2554
- Lansner, A. (2003). Cell assembly dynamics in detailed and abstract attractor models of cortical associative memory. , 122(1), 19–36. doi: 10.1078/1431-7613-00072
- Lansner, A., & Ekeberg, (1989, January). A ONE-LAYER FEEDBACK ARTIFICIAL NEURAL NETWORK WITH A BAYESIAN LEARNING RULE. *International Journal of Neural Systems*, 01(01), 77–87. doi: 10.1142/S0129065789000499
- Lansner, A., Ravichandran, N., & Herman, P. (2025). *Benchmarking hebbian learning rules for associative memory* (No. arXiv:2401.00335). arXiv. doi: 10.48550/arXiv.2401.00335
- Lapique, L. (1907). Recherches quantitatives sur l'excitation électrique des nerfs traitée comme une polarisation. (9), 620–635.
- Lecun, Y., Bottou, L., Bengio, W., & Haffner, P. (1998). Gradient-based learning applied to document recognition. , 86(11), 2278–2324. doi: 10.1109/5.726791
- Lundqvist, M., Compte, A., & Lansner, A. (2010, June). Bistable, Irregular Firing and Population Oscillations in a Modular Attractor Memory Network. *PLoS Computational Biology*, 6(6), e1000803. doi: 10.1371/journal.pcbi.1000803
- Maass, W. (1997). Networks of spiking neurons: The third generation of neural network models. , 10(9), 1659–1671. doi: 10.1016/S0893-6080(97)00011-7
- Mountcastle, V. (1997, April). The columnar organization of the neocortex. *Brain*, 120(4), 701–722. doi: 10.1093/brain/120.4.701

FIGURE B.2: Pre-registration page 2

Bibliography

- Bastos, Andre M. et al. (Nov. 21, 2012). "Canonical Microcircuits for Predictive Coding". In: *Neuron* 76.4, pp. 695–711. ISSN: 0896-6273. DOI: [10.1016/j.neuron.2012.10.038](https://doi.org/10.1016/j.neuron.2012.10.038). URL: <https://www.sciencedirect.com/science/article/pii/S0896627312009592> (visited on 05/20/2026).
- Binzegger, T., R. J. Douglas, and K. a. C. Martin (Oct. 2009). "Topology and dynamics of the canonical circuit of cat V1". In: *Neural Networks: The Official Journal of the International Neural Network Society* 22.8, pp. 1071–1078. ISSN: 1879-2782. DOI: [10.1016/j.neunet.2009.07.011](https://doi.org/10.1016/j.neunet.2009.07.011).
- Brette, Romain and Wulfram Gerstner (Nov. 2005). "Adaptive exponential integrate-and-fire model as an effective description of neuronal activity". In: *Journal of Neurophysiology* 94.5, pp. 3637–3642. ISSN: 0022-3077. DOI: [10.1152/jn.00686.2005](https://doi.org/10.1152/jn.00686.2005).
- Buesing, L. et al. (Nov. 2011). "Neural Dynamics as Sampling: A Model for Stochastic Computation in Recurrent Networks of Spiking Neurons". en. In: *PLoS Computational Biology* 7.11. Ed. by Olaf Sporns, e1002211. ISSN: 1553-7358. DOI: [10.1371/journal.pcbi.1002211](https://doi.org/10.1371/journal.pcbi.1002211).
- Caporale, Natalia and Yang Dan (2008). "Spike timing-dependent plasticity: a Hebbian learning rule". In: *Annual Review of Neuroscience* 31, pp. 25–46. ISSN: 0147-006X. DOI: [10.1146/annurev.neuro.31.060407.125639](https://doi.org/10.1146/annurev.neuro.31.060407.125639).
- Carandini, Matteo and David J. Heeger (Jan. 2012). "Normalization as a canonical neural computation". In: *Nature Reviews Neuroscience* 13.1, pp. 51–62. ISSN: 1471-0048. DOI: [10.1038/nrn3136](https://doi.org/10.1038/nrn3136). URL: <https://www.nature.com/articles/nrn3136> (visited on 05/20/2026).
- Chrysanthis, N. et al. (Aug. 1, 2025). "Short-term plasticity influences episodic memory recall: an interplay of synaptic traces in a spiking neural network model". In: *Scientific Reports* 15.1, p. 28164. ISSN: 2045-2322. DOI: [10.1038/s41598-025-12611-5](https://doi.org/10.1038/s41598-025-12611-5). URL: <https://www.nature.com/articles/s41598-025-12611-5> (visited on 05/20/2026).
- Chrysanthis, Nikolaos, Florian Fiebig, and Anders Lansner (Nov. 5, 2018). *Introducing double bouquet cells into a modular cortical associative memory model*. DOI: [10.1101/462010](https://doi.org/10.1101/462010). URL: <http://biorxiv.org/lookup/doi/10.1101/462010> (visited on 05/20/2026).
- DeFelipe, J. and I. Fariñas (Dec. 1992). "The pyramidal neuron of the cerebral cortex: morphological and chemical characteristics of the synaptic inputs". In: *Progress in Neurobiology* 39.6, pp. 563–607. ISSN: 0301-0082. DOI: [10.1016/0301-0082\(92\)90015-7](https://doi.org/10.1016/0301-0082(92)90015-7).
- Douglas, R. J. et al. (Aug. 18, 1995). "Recurrent excitation in neocortical circuits". In: *Science* 269.5226, pp. 981–985. ISSN: 0036-8075. DOI: [10.1126/science.7638624](https://doi.org/10.1126/science.7638624).
- Fiebig, Florian, Pawel Herman, and Anders Lansner (Mar. 2020). "An Indexing Theory for Working Memory Based on Fast Hebbian Plasticity". In: *eneuro* 7.2, ENEURO.0374–19.2020. ISSN: 2373-2822. DOI: [10.1523/ENEURO.0374-19.2020](https://doi.org/10.1523/ENEURO.0374-19.2020). URL: <https://www.eneuro.org/lookup/doi/10.1523/ENEURO.0374-19.2020> (visited on 05/19/2026).

- Fiebig, Florian and Anders Lansner (Jan. 4, 2017). "A Spiking Working Memory Model Based on Hebbian Short-Term Potentiation". In: *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience* 37.1, pp. 83–96. ISSN: 1529-2401. DOI: [10.1523/JNEUROSCI.1989-16.2016](https://doi.org/10.1523/JNEUROSCI.1989-16.2016).
- Gewaltig, Marc-Oliver and Markus Diesmann (2007). "NEST (NEural Simulation Tool)". In: *Scholarpedia* 2.4, p. 1430. DOI: [10.4249/scholarpedia.1430](https://doi.org/10.4249/scholarpedia.1430).
- Hebb, D.O. (Apr. 11, 1949). *The Organization of Behavior*. 0th ed. Psychology Press. ISBN: 9781410612403. DOI: [10.4324/9781410612403](https://doi.org/10.4324/9781410612403).
- Hopfield, J. J. (Apr. 1982). "Neural networks and physical systems with emergent collective computational abilities." In: *Proceedings of the National Academy of Sciences* 79.8, pp. 2554–2558. ISSN: 0027-8424, 1091-6490. DOI: [10.1073/pnas.79.8.2554](https://doi.org/10.1073/pnas.79.8.2554).
- Hunt, David L. and Pablo E. Castillo (June 2012). "Synaptic plasticity of NMDA receptors: mechanisms and functional implications". In: *Current opinion in neurobiology* 22.3, pp. 496–508. ISSN: 0959-4388. DOI: [10.1016/j.conb.2012.01.007](https://doi.org/10.1016/j.conb.2012.01.007). URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3482462/> (visited on 05/20/2026).
- Kelsom, Corey and Wange Lu (Apr. 23, 2013). "Development and specification of GABAergic cortical interneurons". In: *Cell & Bioscience* 3.1, p. 19. ISSN: 2045-3701. DOI: [10.1186/2045-3701-3-19](https://doi.org/10.1186/2045-3701-3-19).
- Krimer, Leonid S. et al. (Nov. 2005). "Cluster Analysis-Based Physiological Classification and Morphological Properties of Inhibitory Neurons in Layers 2–3 of Monkey Dorsolateral Prefrontal Cortex". In: *Journal of Neurophysiology* 94.5, pp. 3009–3022. ISSN: 0022-3077. DOI: [10.1152/jn.00156.2005](https://doi.org/10.1152/jn.00156.2005). URL: <https://journals.physiology.org/doi/full/10.1152/jn.00156.2005> (visited on 05/20/2026).
- Lansner, A. (2003). "Cell Assembly Dynamics in Detailed and Abstract Attractor Models of Cortical Associative Memory". In: *Theory in Biosciences* 122.1, pp. 19–36. ISSN: 14317613. DOI: [10.1078/1431-7613-00072](https://doi.org/10.1078/1431-7613-00072).
- Lansner, A. and Ö. Ekeberg (Jan. 1989). "A ONE-LAYER FEEDBACK ARTIFICIAL NEURAL NETWORK WITH A BAYESIAN LEARNING RULE". In: *International Journal of Neural Systems* 01.01, pp. 77–87. ISSN: 0129-0657, 1793-6462. DOI: [10.1142/S0129065789000499](https://doi.org/10.1142/S0129065789000499).
- Lansner, A., N. Ravichandran, and P. Herman (Feb. 17, 2025). *Benchmarking Hebbian learning rules for associative memory*. DOI: [10.48550/arXiv.2401.00335](https://doi.org/10.48550/arXiv.2401.00335). arXiv: [2401.00335\[cs\]](https://arxiv.org/abs/2401.00335).
- Lapique, L. (1907). "Recherches quantitatives sur l'excitation électrique des nerfs traitée comme une polarisation". In: *Journal of Physiology and Pathology* 9, pp. 620–635.
- Lecun, Y. et al. (Nov. 1998). "Gradient-based learning applied to document recognition". In: *Proceedings of the IEEE* 86.11, pp. 2278–2324. ISSN: 00189219. DOI: [10.1109/5.726791](https://doi.org/10.1109/5.726791).
- Lisman, John (Mar. 5, 2017). "Glutamatergic synapses are structurally and biochemically complex because of multiple plasticity processes: long-term potentiation, long-term depression, short-term potentiation and scaling". In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 372.1715, p. 20160260. ISSN: 0962-8436. DOI: [10.1098/rstb.2016.0260](https://doi.org/10.1098/rstb.2016.0260). URL: <https://doi.org/10.1098/rstb.2016.0260> (visited on 05/20/2026).
- Lodato, Simona and Paola Arlotta (2015). "Generating neuronal diversity in the mammalian cerebral cortex". In: *Annual Review of Cell and Developmental Biology* 31, pp. 699–720. ISSN: 1530-8995. DOI: [10.1146/annurev-cellbio-100814-125353](https://doi.org/10.1146/annurev-cellbio-100814-125353).

- Lundqvist, Mikael (2013). "Oscillations and spike statistics in biophysical attractor networks". Doctoral Thesis. Sweden: Stockholm University.
- Lundqvist, Mikael, Pawel Herman, and Anders Lansner (Oct. 2011). "Theta and gamma power increases and alpha/beta power decreases with memory load in an attractor network model". In: *Journal of Cognitive Neuroscience* 23.10, pp. 3008–3020. ISSN: 1530-8898. DOI: [10.1162/jocn_a_00029](https://doi.org/10.1162/jocn_a_00029).
- Mayorquin, Ramon Heberto Martinez (2022). "Sequence learning in the Bayesian Confidence Propagation Neural Network". PhD thesis. Sweden: KTH Royal Institute of Technology.
- McBain, Chris J (Feb. 15, 2012). "Cortical inhibitory neuron basket cells: from circuit function to disruption". In: *The Journal of Physiology* 590 (Pt 4), p. 667. ISSN: 0022-3751. DOI: [10.1113/jphysiol.2012.227967](https://doi.org/10.1113/jphysiol.2012.227967). URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3381300/> (visited on 05/20/2026).
- Meng, Pinchao et al. (2022). "ODE-RU: a dynamical system view on recurrent neural networks". In: *Electronic Research Archive* 30.1. Cc_license_type: cc_by Primary_atype: Electronic Research Archive Subject_term: Research article Subject_term_id: Research article, pp. 257–271. ISSN: 2688-1594. DOI: [10.3934/era.2022014](https://doi.org/10.3934/era.2022014). URL: <http://www.aimspress.com/article/doi/10.3934/era.2022014> (visited on 05/20/2026).
- Mountcastle, V. (Apr. 1997). "The columnar organization of the neocortex". In: *Brain* 120.4, pp. 701–722. ISSN: 14602156. DOI: [10.1093/brain/120.4.701](https://doi.org/10.1093/brain/120.4.701).
- Mountcastle, V. B. (July 1957). "Modality and topographic properties of single neurons of cat's somatic sensory cortex". In: *Journal of Neurophysiology* 20.4, pp. 408–434. ISSN: 0022-3077. DOI: [10.1152/jn.1957.20.4.408](https://doi.org/10.1152/jn.1957.20.4.408).
- Ravichandran, N., A. Lansner, and P. Herman (Sept. 19, 2024). "Spiking representation learning for associative memories". In: *Frontiers in Neuroscience* 18, p. 1439414. ISSN: 1662-453X. DOI: [10.3389/fnins.2024.1439414](https://doi.org/10.3389/fnins.2024.1439414). arXiv: [2406.03054](https://arxiv.org/abs/2406.03054)[cs].
- (Apr. 14, 2025). "Unsupervised representation learning with Hebbian synaptic and structural plasticity in brain-like feedforward neural networks". In: *Neurocomputing* 626, p. 129440. ISSN: 0925-2312. DOI: [10.1016/j.neucom.2025.129440](https://doi.org/10.1016/j.neucom.2025.129440).
- Sandberg, Anders, Jesper Tegner, and Anders Lansner (Dec. 1, 2003). "A working memory model based on fast Hebbian learning". In: *Network (Bristol, England)* 14, pp. 789–802. DOI: [10.1088/0954-898X/14/4/309](https://doi.org/10.1088/0954-898X/14/4/309).
- Seidl, Armin H. (Sept. 12, 2014). "Regulation of Conduction Time along Axons". In: *Neuroscience* 0, pp. 126–134. ISSN: 0306-4522. DOI: [10.1016/j.neuroscience.2013.06.047](https://doi.org/10.1016/j.neuroscience.2013.06.047). URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3849146/> (visited on 05/20/2026).
- Stimberg, M., R. Brette, and D. Goodman (Aug. 20, 2019). "Brian 2, an intuitive and efficient neural simulator". In: *eLife* 8, e47314. ISSN: 2050-084X. DOI: [10.7554/eLife.47314](https://doi.org/10.7554/eLife.47314).
- Sukenik, N. et al. (Mar. 23, 2021). "Neuronal circuits overcome imbalance in excitation and inhibition by adjusting connection numbers". In: *Proceedings of the National Academy of Sciences* 118.12, e2018459118. ISSN: 0027-8424, 1091-6490. DOI: [10.1073/pnas.2018459118](https://doi.org/10.1073/pnas.2018459118).
- Thomson, Alex M. and A. Peter Bannister (Jan. 2003). "Interlaminar connections in the neocortex". In: *Cerebral Cortex* 13.1, pp. 5–14. ISSN: 1047-3211. DOI: [10.1093/cercor/13.1.5](https://doi.org/10.1093/cercor/13.1.5).
- Thomson, Alex M. et al. (Sept. 1, 2002). "Synaptic Connections and Small Circuits Involving Excitatory and Inhibitory Neurons in Layers 2–5 of Adult Rat and Cat Neocortex: Triple Intracellular Recordings and Biocytin Labelling In Vitro". In: *Cerebral Cortex* 12.9, pp. 936–953. ISSN: 1047-3211. DOI: [10.1093/cercor/12.9.936](https://doi.org/10.1093/cercor/12.9.936).

- 9.936. URL: <https://dx.doi.org/10.1093/cercor/12.9.936> (visited on 03/03/2026).
- Tsien, J. Z. (Apr. 2000). "Linking Hebb's coincidence-detection to memory formation". In: *Current Opinion in Neurobiology* 10.2, pp. 266–273. ISSN: 0959-4388. DOI: [10.1016/s0959-4388\(00\)00070-2](https://doi.org/10.1016/s0959-4388(00)00070-2).
- Tsodyks, Misha V. and Henry Markram (Jan. 21, 1997). "The neural code between neocortical pyramidal neurons depends on neurotransmitter release probability". In: *Proceedings of the National Academy of Sciences* 94.2, pp. 719–723. DOI: [10.1073/pnas.94.2.719](https://doi.org/10.1073/pnas.94.2.719). URL: <https://www.pnas.org/doi/10.1073/pnas.94.2.719> (visited on 02/20/2026).
- Tully, Philip J., Matthias H. Hennig, and Anders Lansner (Apr. 8, 2014). "Synaptic and nonsynaptic plasticity approximating probabilistic inference". In: *Frontiers in Synaptic Neuroscience* 6. ISSN: 1663-3563. DOI: [10.3389/fnsyn.2014.00008](https://doi.org/10.3389/fnsyn.2014.00008). URL: <https://www.frontiersin.org/journals/synaptic-neuroscience/articles/10.3389/fnsyn.2014.00008/full> (visited on 05/20/2026).
- Vries-Gao, Alex de (Jan. 9, 2026). "The carbon and water footprints of data centers and what this could mean for artificial intelligence". In: *Patterns* 7.1, p. 101430. ISSN: 2666-3899. DOI: [10.1016/j.patter.2025.101430](https://doi.org/10.1016/j.patter.2025.101430). URL: <https://www.sciencedirect.com/science/article/pii/S2666389925002788> (visited on 05/20/2026).
- Werbos, Paul (Nov. 1, 1990). "Backpropagation through time: what it does and how to do it". In: *Proceedings of the IEEE* 78, pp. 1550–1560. DOI: [10.1109/5.58337](https://doi.org/10.1109/5.58337).